IGRF-13: evaluation of candidate models

Authors: William Brown, Ciaran Beggan, Susan Macmillan and Grace Cox
British Geological Survey
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Definitive Geomagnetic Reference Field (DGRF) for 2015.0

There are eleven candidate models for the DGRF-2015:
A BGS
C CU/NCEI
D DTU
E GFZ
F IPGP
G ISTerre
H IZMIRAN
L NASA/GSFC
M Potsdam
N Spain
O Strasbourg/EOST

All are based on Swarm Level 1b data and many include ground-based observatory data. Thus, as expected, most have very similar Gauss coefficients.

A simple RMS difference of coefficients, in Figure 1 and Table 1, reveals that most are within 4 nT RMS of each other and the median and mean models. The exceptions are the agreement of the Strasbourg/EOST, ISTerre and IZMIRAN DGRF candidates which have slightly higher differences of between 6 and 8 nT.

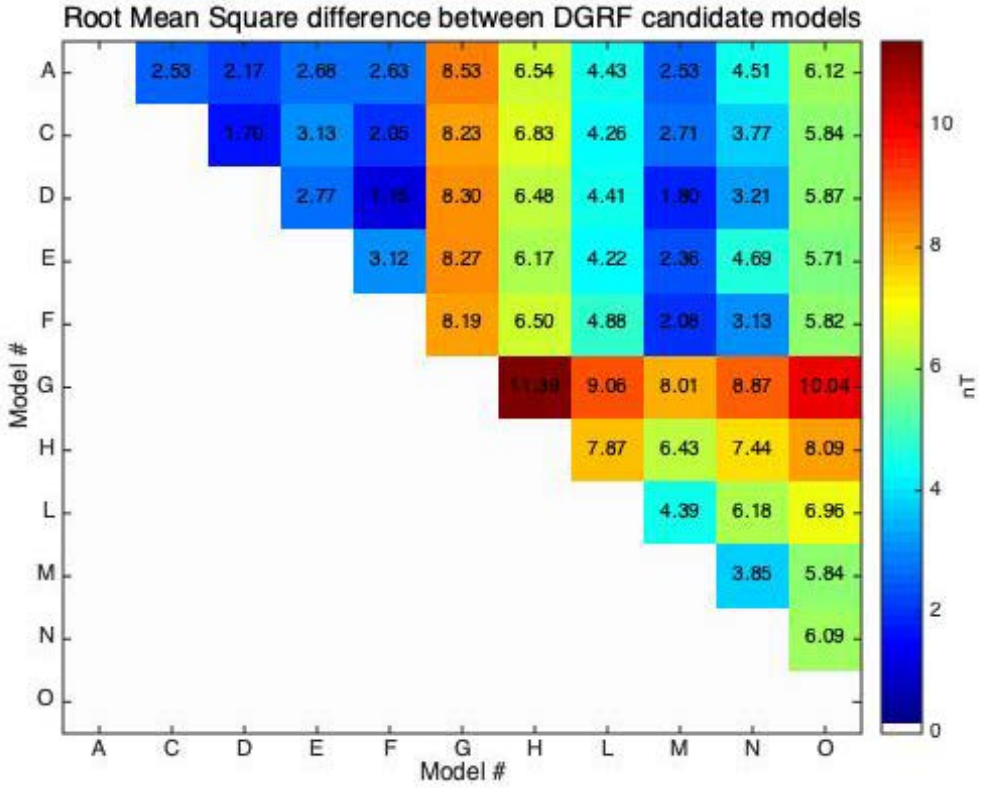


Figure 1: RMS differences between DGRF candidates

	A	C	D	E	F	G	H	L	M	N	O	Mean	Median
A		2.53	2.17	2.68	2.63	8.53	6.54	4.43	2.53	4.51	6.12	2.25	2.02
C	2.53		1.70	3.13	2.05	8.23	6.83	4.26	2.71	3.77	5.84	1.96	1.74
D	2.17	1.70		2.77	1.15	8.30	6.48	4.41	1.80	3.21	5.87	1.52	0.93
E	2.68	3.13	2.77		3.12	8.27	6.17	4.22	2.36	4.69	5.71	2.12	2.31
F	2.63	2.05	1.15	3.12		8.19	6.50	4.88	2.08	3.13	5.82	1.76	1.25
G	8.53	8.23	8.30	8.27	8.19		11.39	9.06	8.01	8.87	10.04	7.57	8.06
H	6.54	6.83	6.48	6.17	6.50	11.39		7.87	6.43	7.44	8.09	6.01	6.23
L	4.43	4.26	4.41	4.22	4.88	9.06	7.87		4.39	6.18	6.96	4.04	4.32
M	2.53	2.71	1.80	2.36	2.08	8.01	6.43	4.39		3.85	5.84	1.69	1.58
N	4.51	3.77	3.21	4.69	3.13	8.87	7.44	6.18	3.85		6.09	3.45	3.22
O	6.12	5.84	5.87	5.71	5.82	10.04	8.09	6.96	5.84	6.09		5.12	5.42
Mean Diff	4.27	4.11	3.79	4.31	3.96	8.89	7.37	5.66	4.00	5.18	6.64	3.41	3.37

Table 1: RMS differences between candidate models (in nT).

A Lowes-Mauesberger plot again shows the similarities between the candidates (Figure 2).

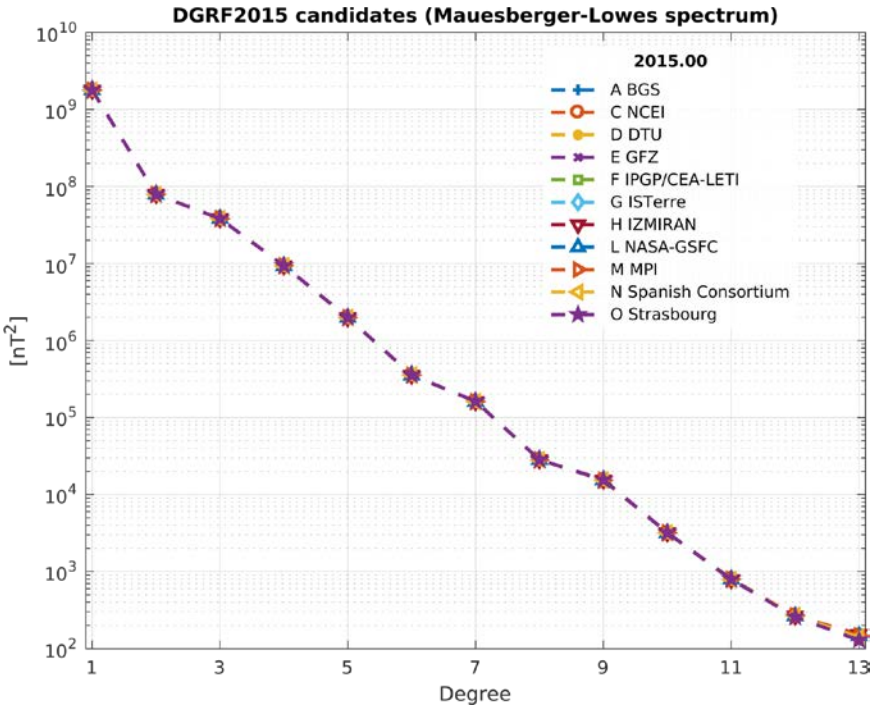


Figure 2: Lowes-Mauersberger spectrum for the 11 DGRF-2015 candidates

As a baseline, we use the BGS candidate to make comparisons to the other candidates. The following plots show the comparison between the BGS candidate model and the other candidates via the mean square difference (Figure 3) and degree correlation (Figure 4), with coefficient differences as triangle plots (
Figure 5) and spatial maps (
Figure 6,
Figure 7 and Figure 8).

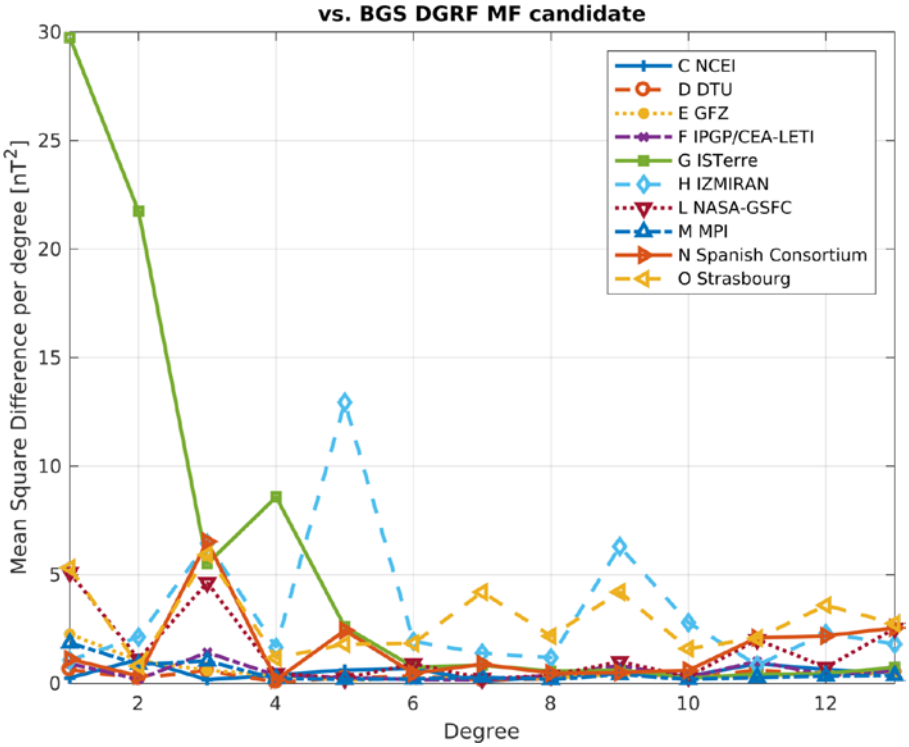


Figure 3: Mean square difference per degree between BGS and the other DGRF-2015 candidates

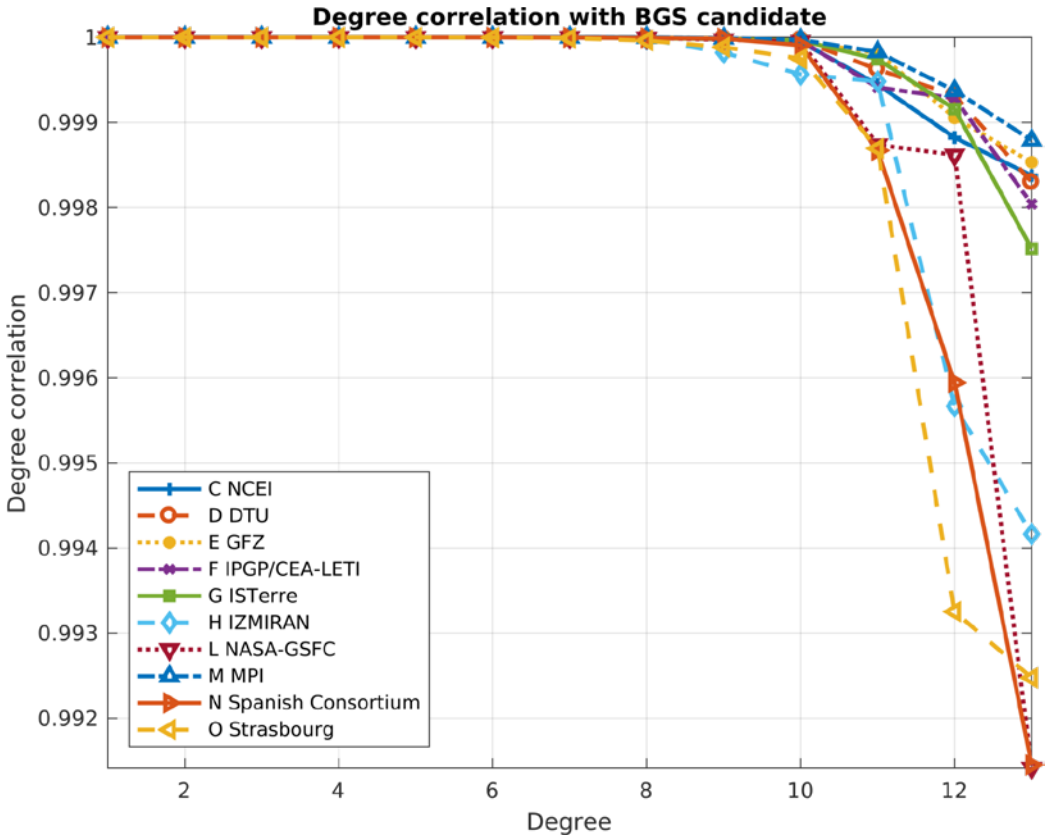


Figure 4: Degree correlation between BGS and the other DGRF-2015 candidates

The MS difference per degree plot shows degree 1 and 2 coefficients of the ISTerre model are largest, while degree 5 and 7 of the IZMIRAN model are greatest. Degree correlation is high at over 0.999 for all model comparisons to degree 10.

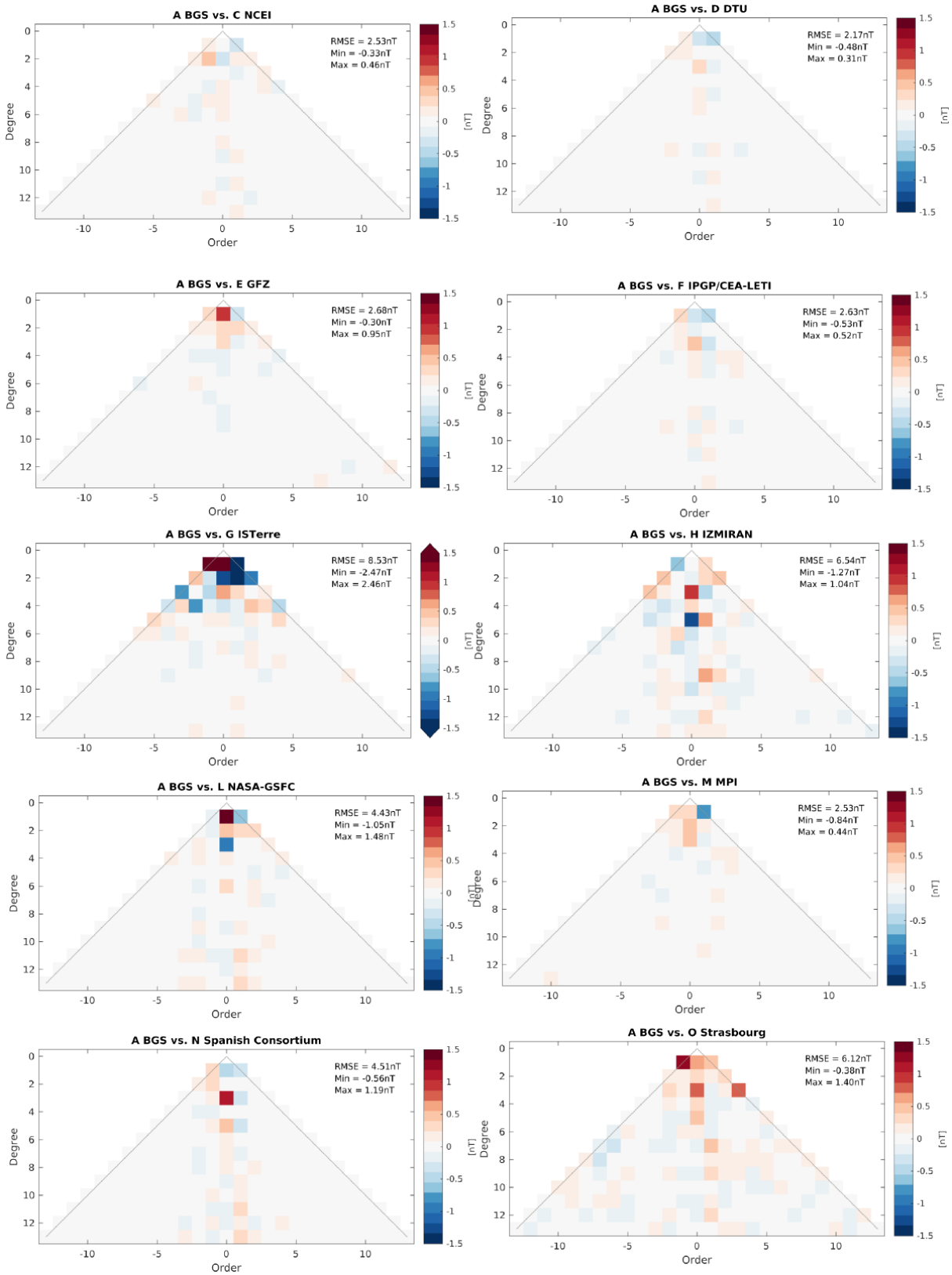


Figure 5: Triangle plots of coefficient differences between BGS and the other DGRF-2015 candidates

From the triangle plots we see the Spanish consortium differs most at degree 3, as does the NASA/GSFC model.

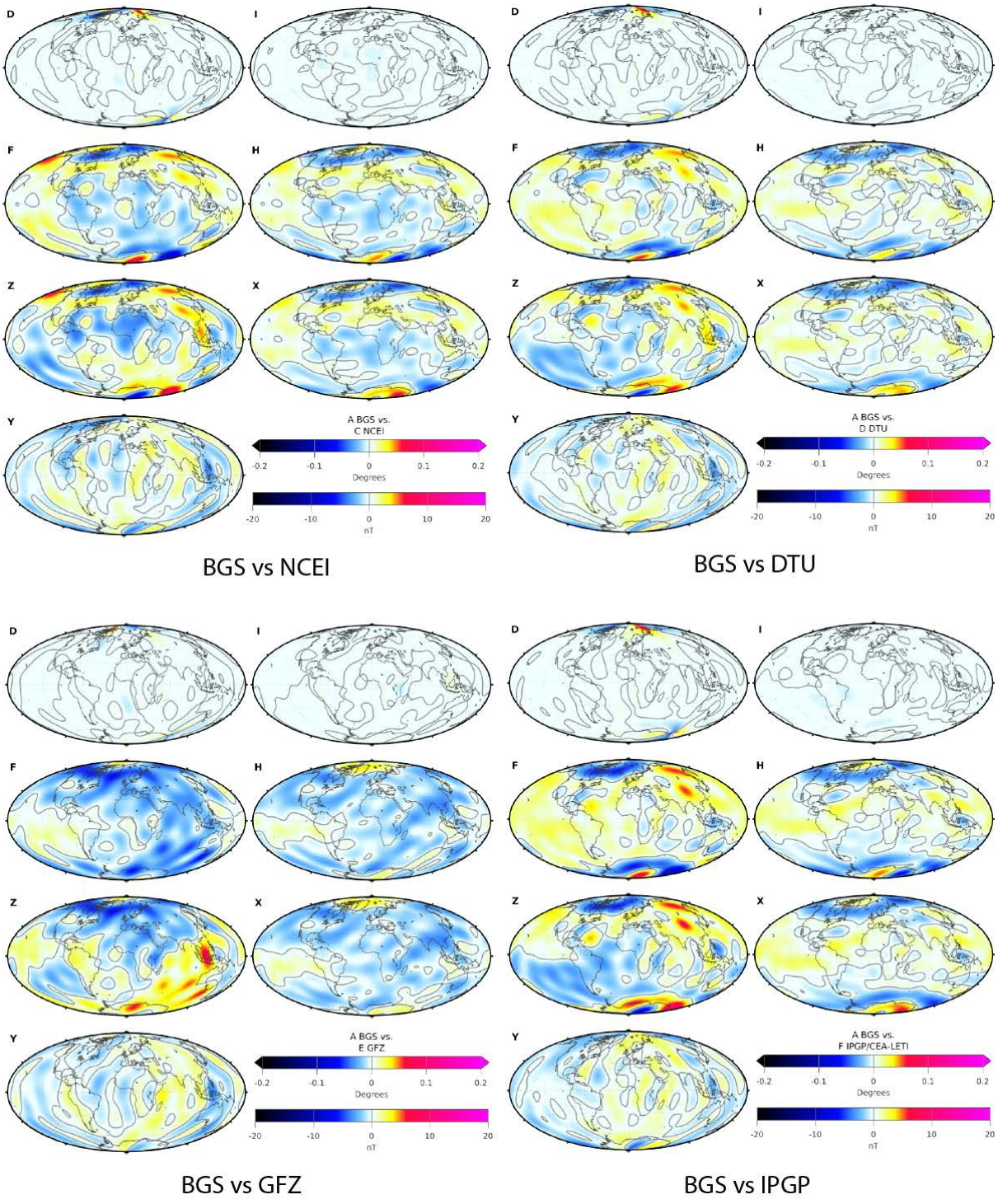


Figure 6: Map comparisons between the BGS DGRF candidate and NCEI, DTU, GFZ and IPGP

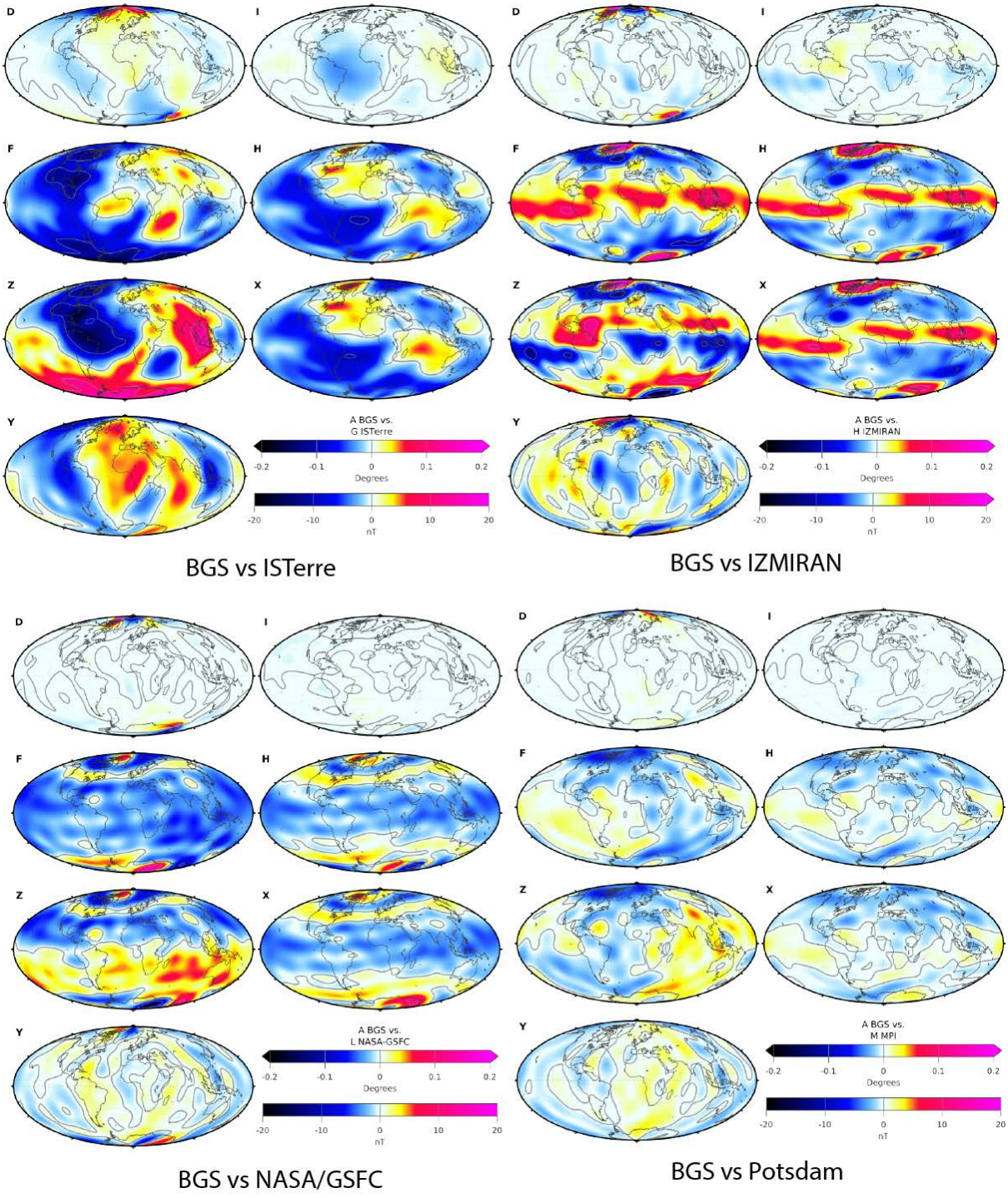


Figure 7: Map comparisons between the BGS DGRF candidate and ISTerre, IZMIRAN, NASA/GSFC and Potsdam

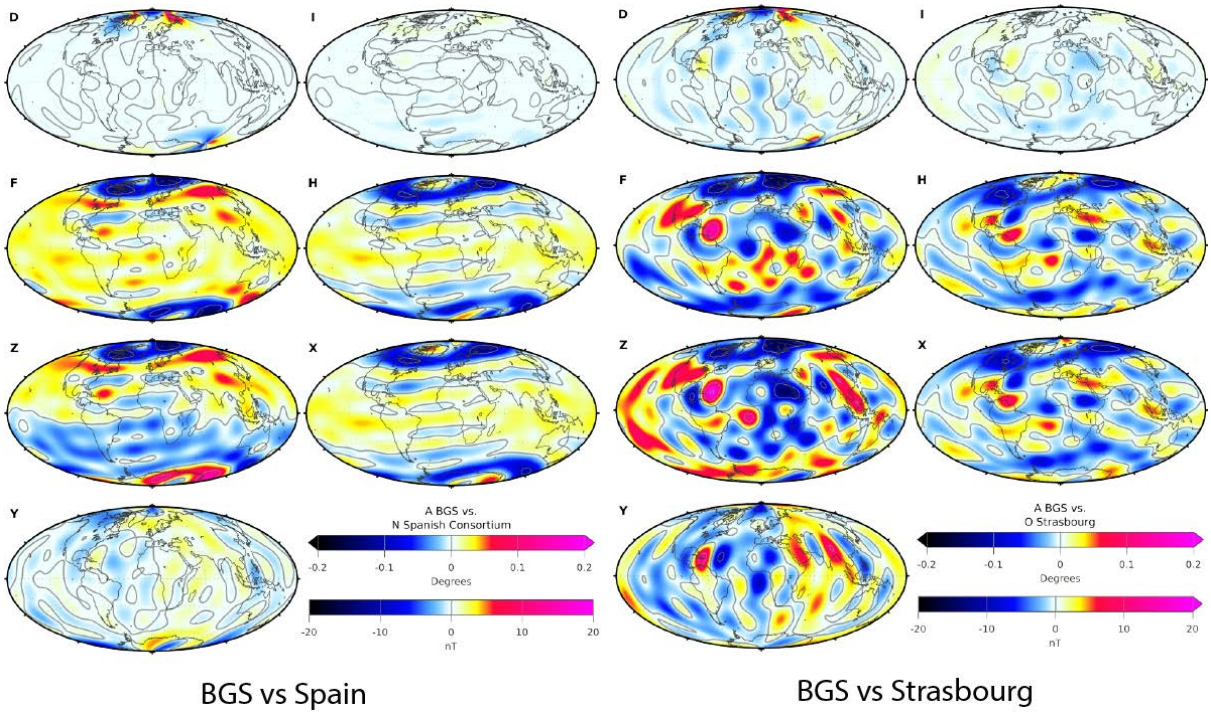


Figure 8: Map comparisons between the BGS DGRF candidate and the Spanish Consortium and Strosbourg/EOST

The mapped differences between the BGS, IZMIRAN and Spanish consortium model show differences at the auroral regions with a strong zonal pattern, while the ISTerre and Strasbourg model differences are more complex.

The other models are quite similar to the BGS (and each other) with differences in Z typically less than 10 nT at any point on the globe (excluding the auroral regions).

International Geomagnetic Reference Field (IGRF) for 2020.0

There are twelve candidate models for the IGRF-2020.
A BGS
B CSES
C CU/NCEI
D DTU
E GFZ
F IPGP
G ISTerre
H IZMIRAN
L NASA/GSFC
M Potsdam
N Spanish consortium
O Strasbourg/EOST

Ten are based on Swarm Level 1b data and again many include ground-based observatory data. The CSES model is based on data solely from the Chinese Seismology and Electromagnetism Satellite, which does not fly in a polar orbit.

Many use some simple form of extrapolation from some point in 2019 to extend their main field from the end of the available satellite data to 2020.0. This leads to a wider variation of the RMS differences between the candidates' Gauss coefficients than the DGRF candidates, as shown in Figure 9.

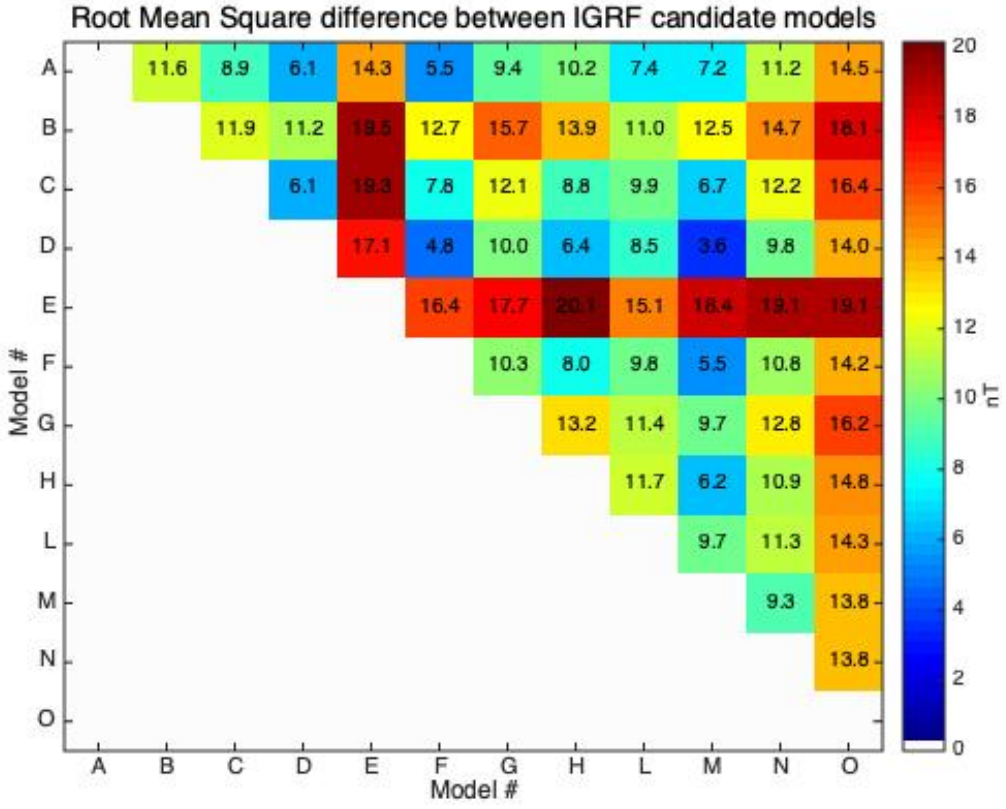


Figure 9:RMS Difference between IGRF candidates

	A	B	C	D	E	F	G	H	L	M	N	O	Mean	Median
A		11.63	8.87	6.07	14.30	5.53	9.35	10.23	7.41	7.23	11.19	14.54	4.61	5.09
B	11.63		11.89	11.16	19.52	12.69	15.72	13.88	10.99	12.49	14.74	18.10	10.56	11.14
C	8.87	11.89		6.12	19.28	7.82	12.11	8.80	9.87	6.68	12.23	16.38	7.12	6.56
D	6.07	11.16	6.12		17.08	4.76	9.98	6.36	8.48	3.63	9.83	13.95	3.69	2.48
E	14.30	19.52	19.28	17.08		16.38	17.66	20.15	15.07	18.38	19.10	19.06	14.88	15.96
F	5.53	12.69	7.82	4.76	16.38		10.30	8.02	9.85	5.47	10.78	14.19	4.93	4.18
G	9.35	15.72	12.11	9.98	17.66	10.30		13.22	11.36	9.71	12.79	16.22	8.93	8.93
H	10.23	13.88	8.80	6.36	20.15	8.02	13.22		11.68	6.23	10.92	14.83	7.69	6.66
L	7.41	10.99	9.87	8.48	15.07	9.85	11.36	11.68		9.68	11.32	14.31	6.42	7.62
M	7.23	12.49	6.68	3.63	18.38	5.47	9.71	6.23	9.68		9.26	13.77	4.75	3.40
N	11.19	14.74	12.23	9.83	19.10	10.78	12.79	10.92	11.32	9.26		13.84	8.62	8.93
O	14.54	18.10	16.38	13.95	19.06	14.19	16.22	14.83	14.31	13.77	13.84		12.14	12.84
Mean Diff	9.67	13.89	10.91	8.86	17.81	9.62	12.58	11.30	10.91	9.32	12.36	15.38	7.86	7.82

Table 2: RMS differences between IGRF-2020 candidate models

From the 66 inter-comparisons, 24 have an RMS difference below 10 nT. Thirty-six have a difference less than 12 nT. There is a reasonably uniform distribution of the differences, which suggests no clear 'best' population of models. However, the CEA, Strasbourg/EOST and GFZ candidates are most different from the rest of the population with consistently higher differences compared to other candidates. This is likely the result of their choice for extrapolation method from the end of the input data in 2019.

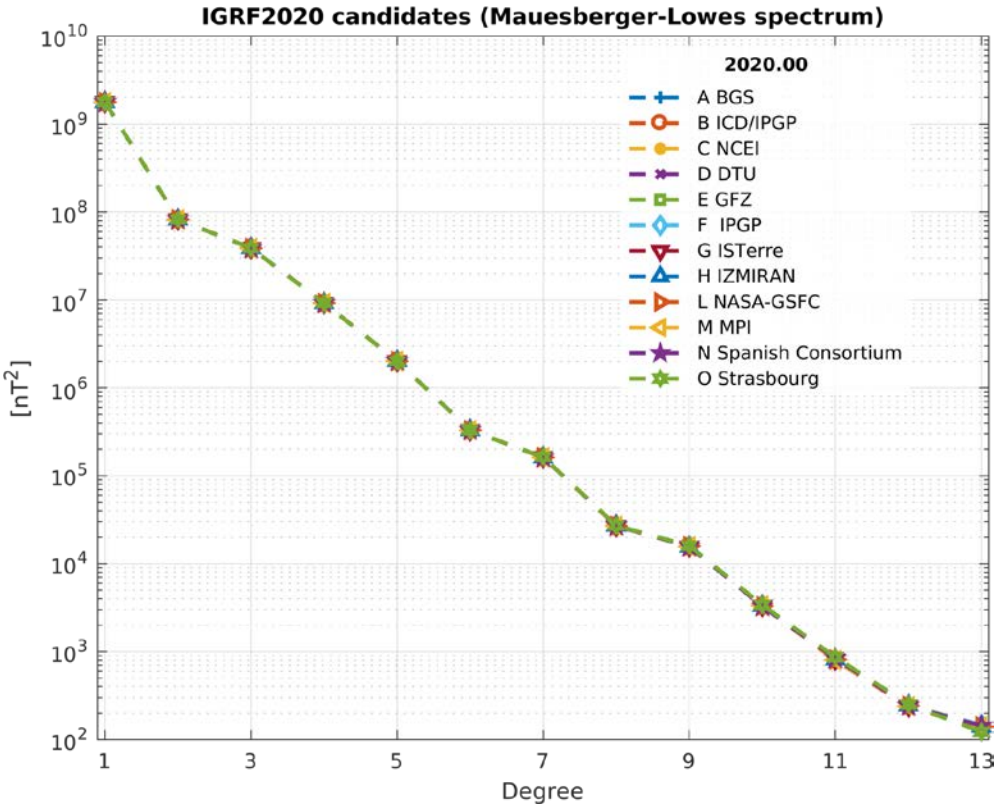


Figure 10: Lowes-Maursberger spectrum for the 12 IGRF-2020 candidates

Again as a baseline, we use the BGS candidate to make comparisons to the other candidates. The following plots show the comparison between the BGS candidate model and the other candidates via the mean square difference and degree correlation, with coefficient differences as triangle plots and spatial maps.

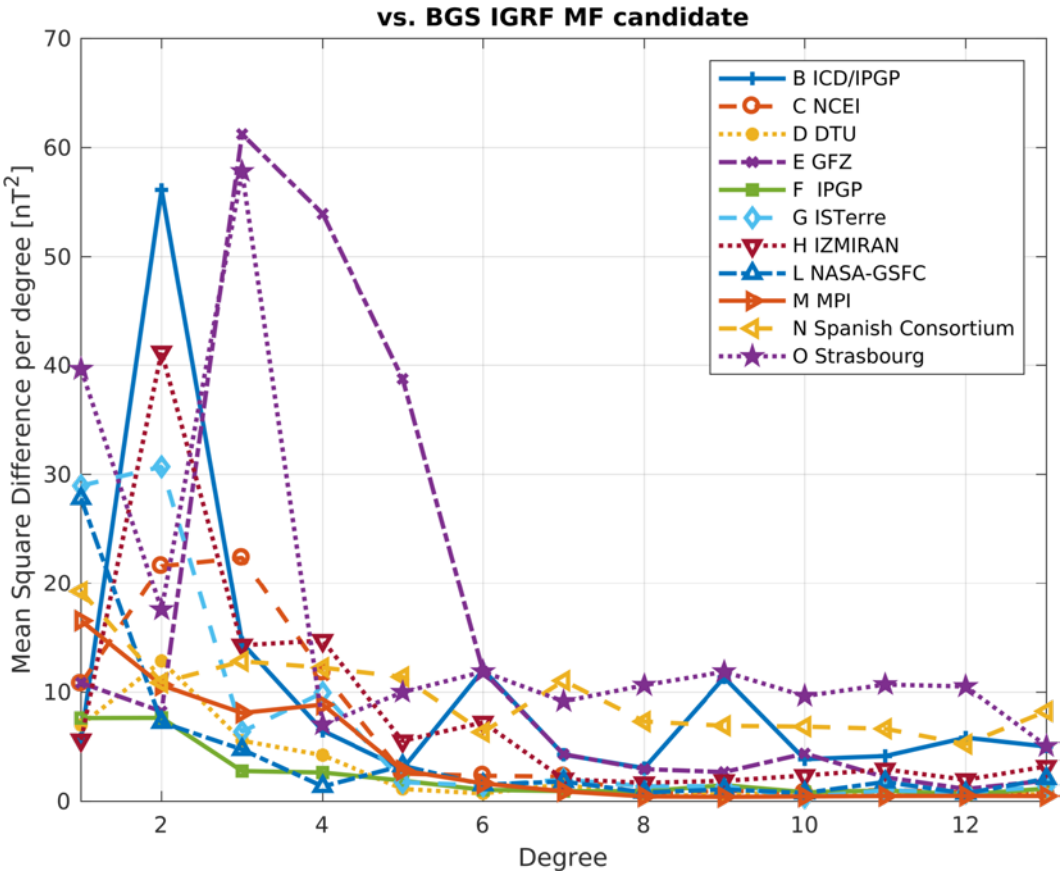


Figure 11: Mean square difference per degree between BGS and the other IGRF-2020 candidates

There is a much wider variation in the mean square differences per degree for the IGRF-2020 candidates. The CEA model differs in degree 2, GFZ in the degrees 3, 4 and 5, IZMIRAN mostly in degree 2, and Strasbourg in degrees 1 and 3.

The degree correlation is above 0.99 until degree 11 for all candidates.

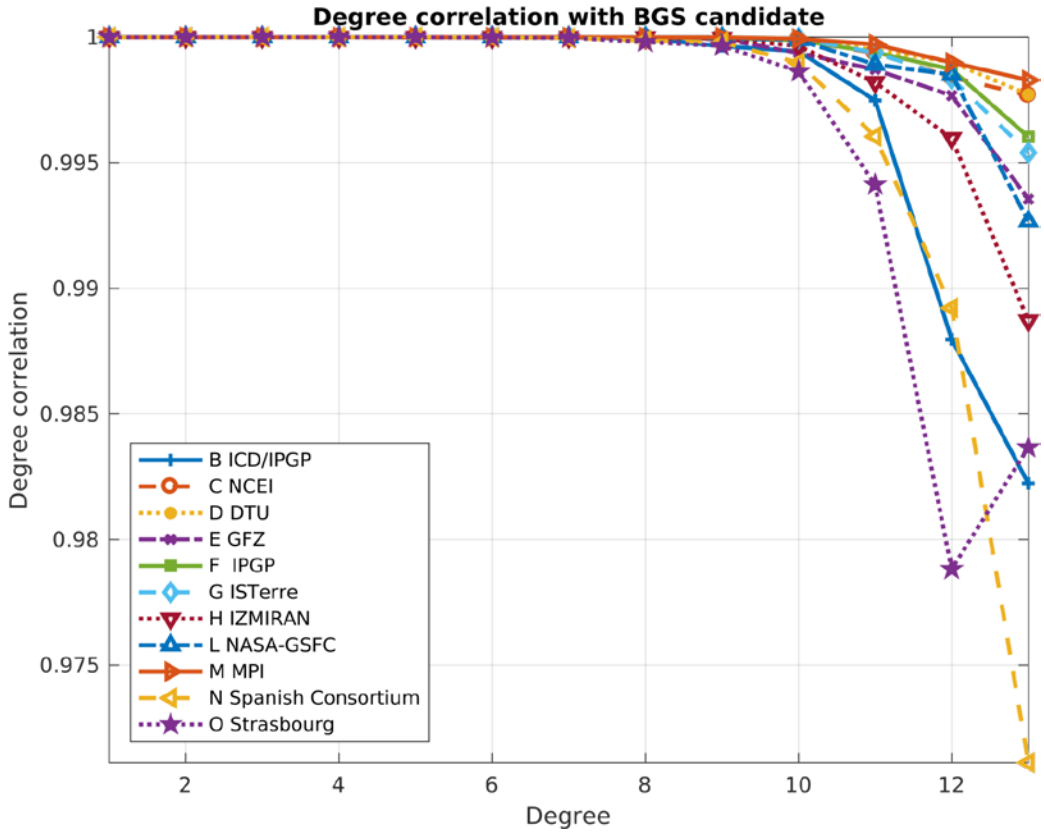


Figure 12: Degree correlation between BGS and the other IGRF-2020 candidates

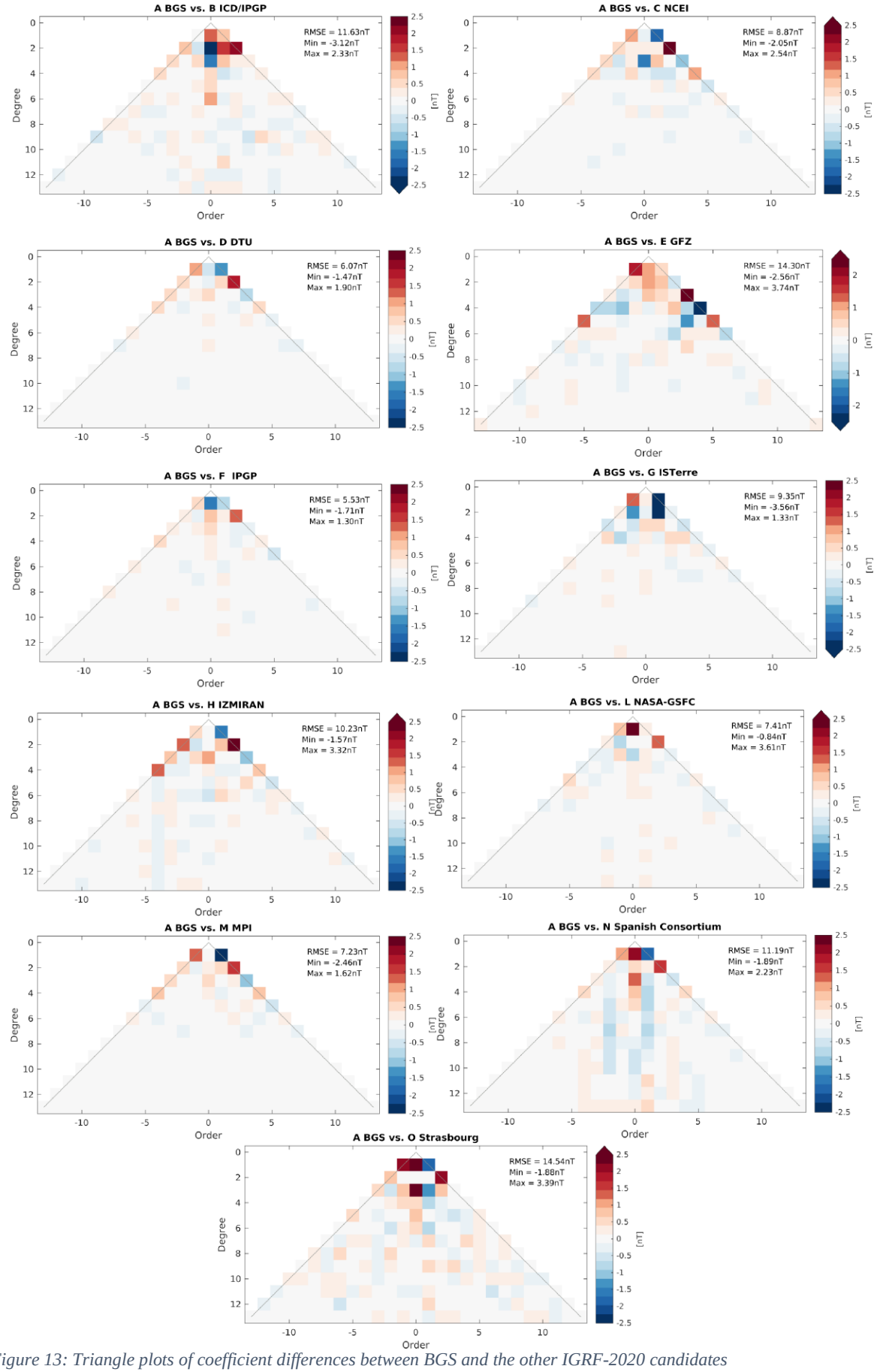


Figure 13: Triangle plots of coefficient differences between BGS and the other IGRF-2020 candidates

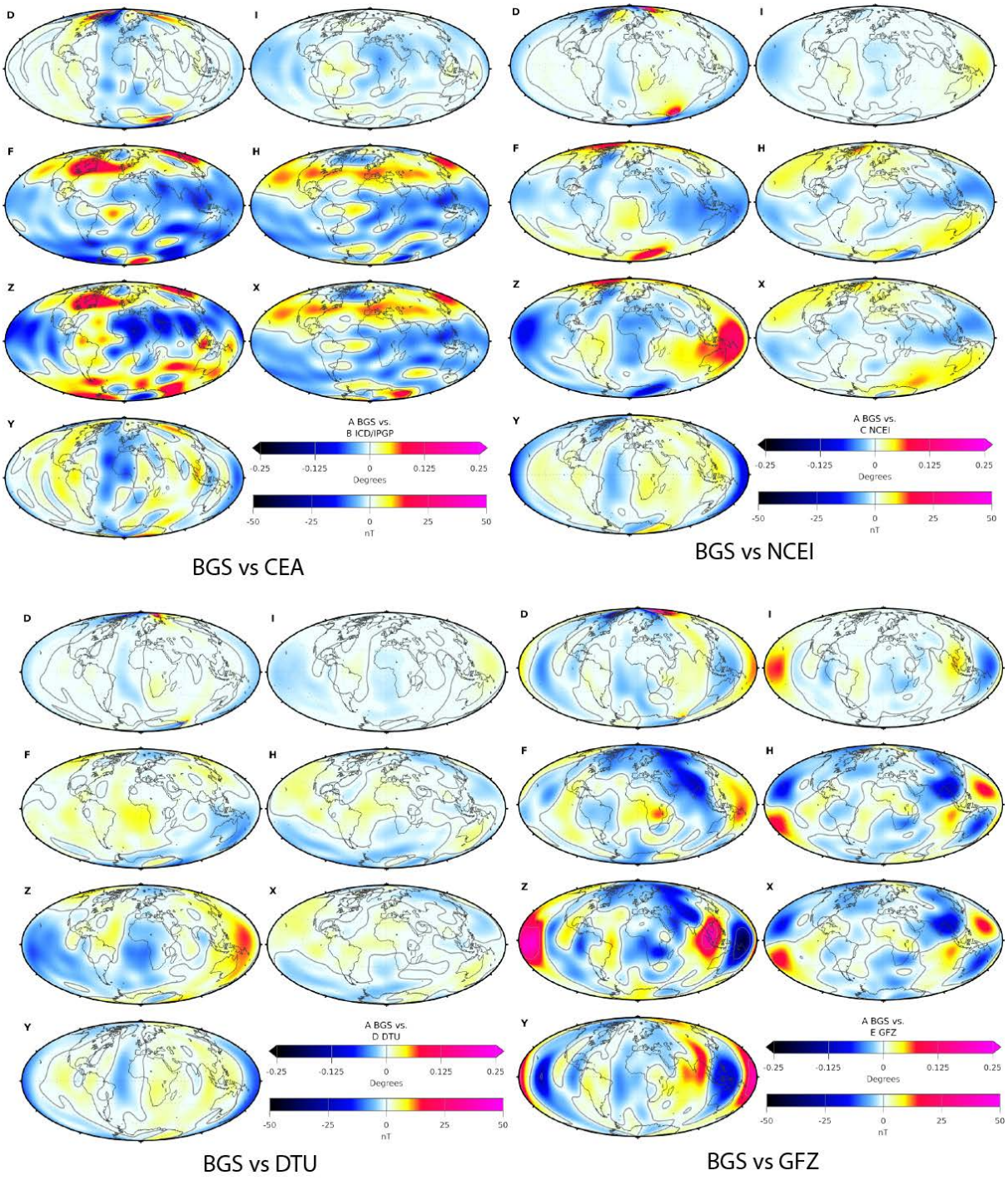


Figure 14: Map comparisons between the BGS IGRF candidate and CEA, NCEI, DTU and GFZ

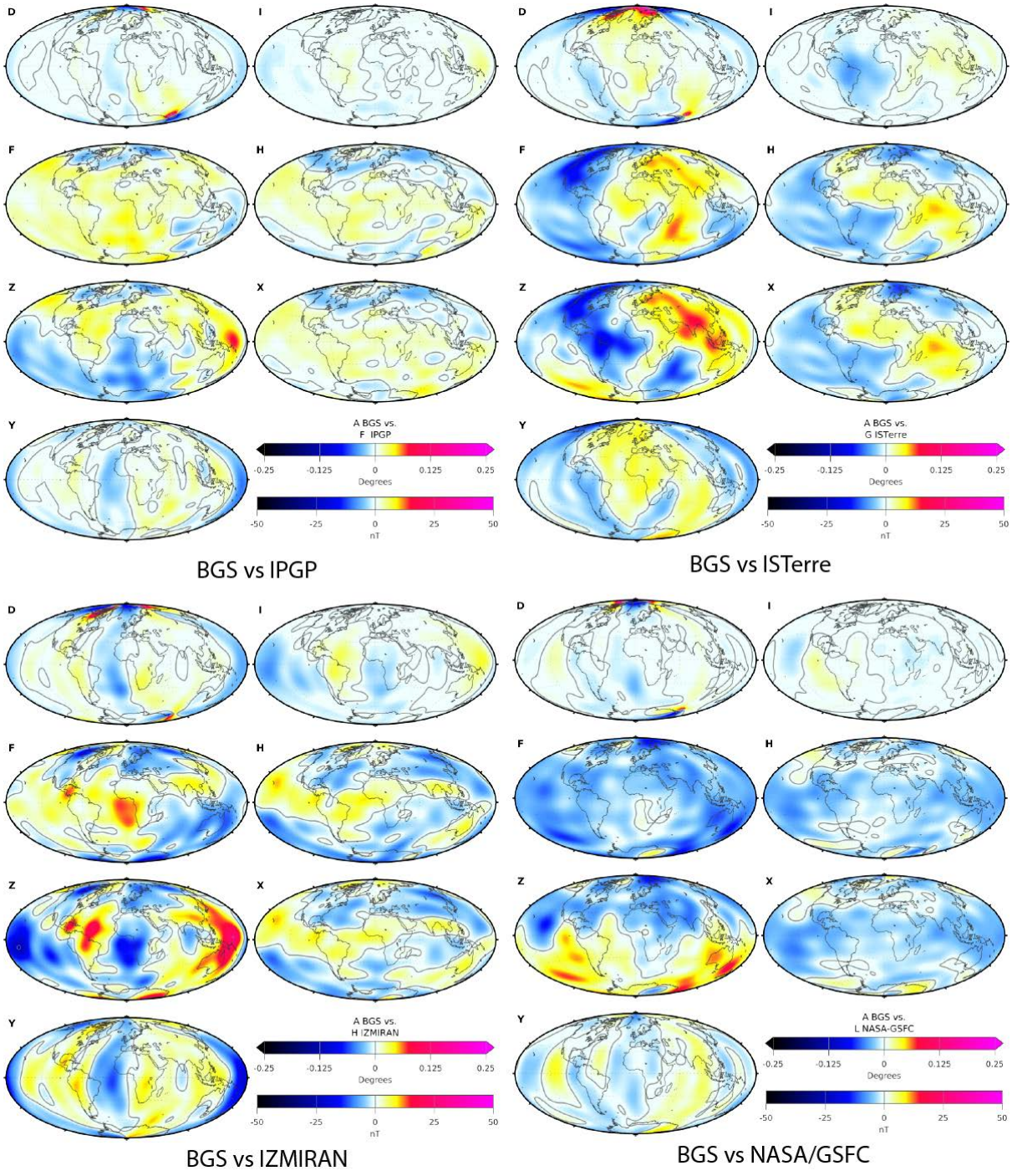


Figure 15: Map comparisons between the BGS IGRF candidate and IPGP, ISTerre, IZMIRAN and NASA/GSFC

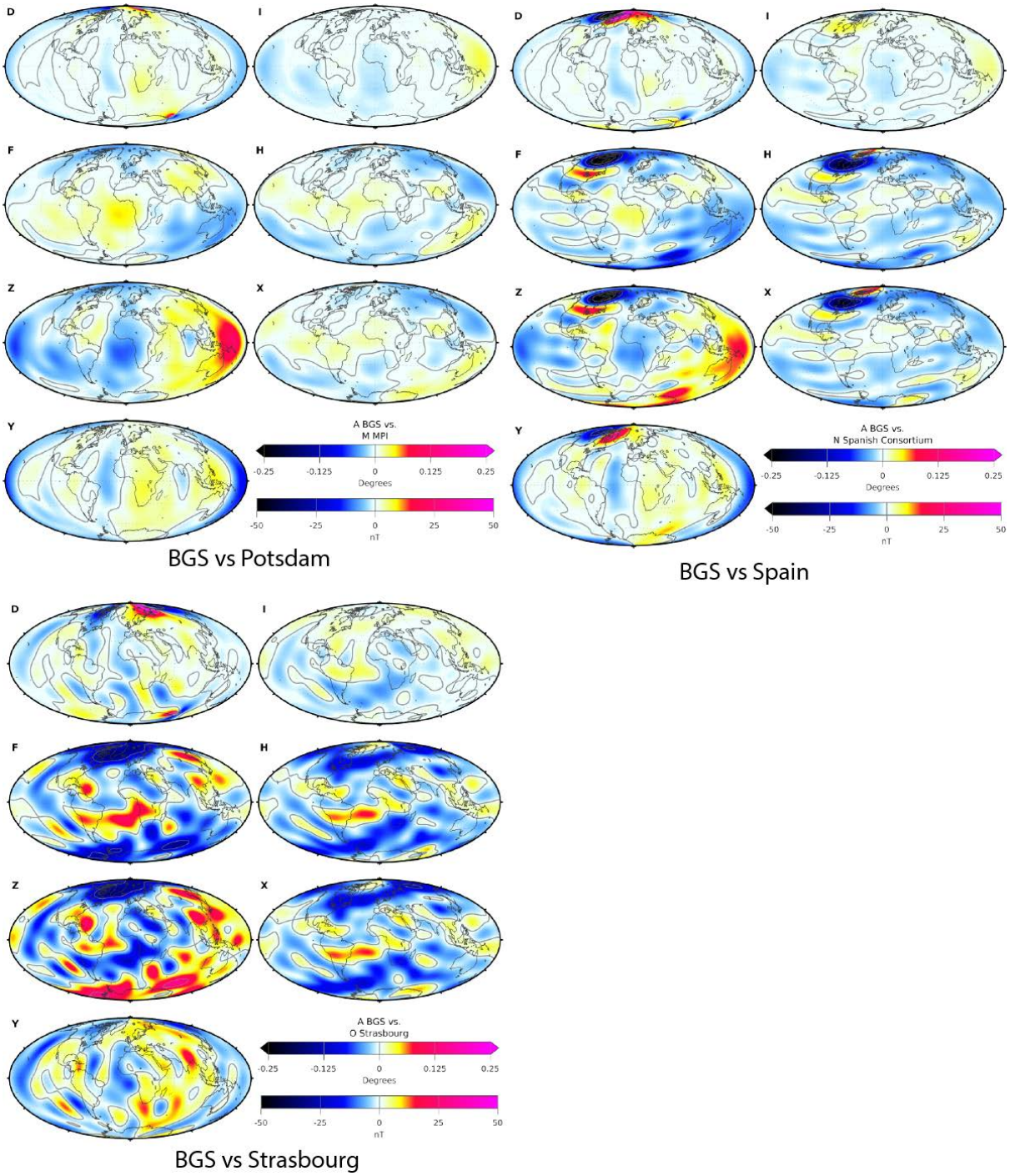


Figure 16: Map comparisons between the BGS IGRF candidate and Potsdam, Spanish consortium and Strasbourg/EOST

The mapped differences again show the largest variations are with the CEA, GFZ and Strasbourg/EOST candidates. Distinct zonal differences are seen for the Spanish consortium model as with the DGRF candidate. Of note is the prevalence in most of these comparison maps (best seen in Z, see CU/NCEI, DTU, GZF, IPGP, IZMIRAN, Potsdam, Spanish consortium, and compare to predicted SV maps in Figure 24 to Figure 27) of the pattern of recent SV, with largest discrepancies where SV has been most intense (e.g. around +/-170 degrees longitude). This suggests that primarily the magnitude of predicted SV used to extrapolate each candidate to 2020 differs. Those candidate model differences that are not solely dominated by this pattern are CEA, ISTerre, IZMIRAN, NASA-GSFC, Spanish consortium and Strasbourg.

Secular Variation (SV) for 2020.0-2025.0

There are a set of fourteen candidate models for the secular variation of the field from 2020.0 to 2025.0.

A BGS
C CU/NCEI
D DTU
E GFZ
F IPGP
G ISTerre
H IZMIRAN_2020
I Japan
J Leeds
K Max Planck
L NASA/GSFC
M Potsdam
N Spain
O Strasbourg

There are a variety of approaches for forecasting secular variation. BGS use core flows with steady flow and acceleration, while CU/NCEI, DTU and GFZ use extrapolation of the secular variation from the previous year(s). The Spanish team have made predictions in a similar manner, but produced from an ensemble of field models built with a bootstrap approach from subsets of observations. The IPGP, Japan, Potsdam, Max Planck and NASA/GSFC candidates use assimilation of ground and/or satellite data with geodynamo model outputs to form an estimate of secular variation. ISTerre uses an ensemble of models from a stochastic prior and predicts with a Best Linear Unbiased Estimator. Leeds have used core flow combined with magnetic diffusion for their estimates. Strasbourg/EOST have employed a singular spectral analysis of sixty years of SV to provide their candidate values. IZMIRAN produce a spectral fit to each Gauss coefficient.

The candidates offer a very diverse set of approaches to compute SV over the period of 2020.0 to 2025.0 which produces a wide variation in the comparison between the candidates, as seen in Figure 17.

The candidate differences fall into two fairly distinct categories – those that match each other to within 12 nT/yr and those whose differences are greater than 20 nT/yr. Figure 18 shows the histogram of RMS differences. The candidates most different from the rest of the population are the NASA/GSFC, Strasbourg/EOST, and Japanese models.

The Japanese model in particular has a very small dipole SV coefficient of 1.9 nT/yr compared to 6-7 nT/yr for the other models. The Strasbourg/EOST model has a high g_1^1 coefficient of 11.5 nT/yr compared to 7-8 nT/yr in other candidates. These coefficients contribute largely to the differences between these two and the others.

Finally, the NASA/GSFC model has very different degree 2 zonal coefficients compared to the other candidates and so shows strong variation.

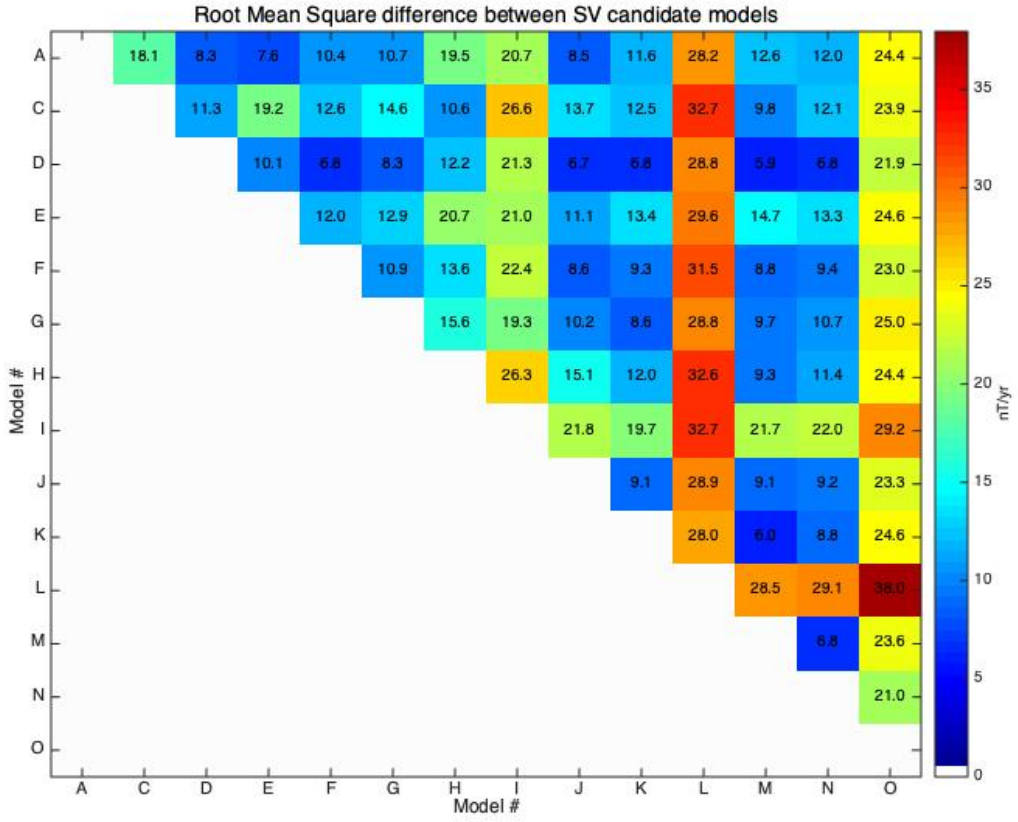


Figure 17: RMS difference between SV candidates

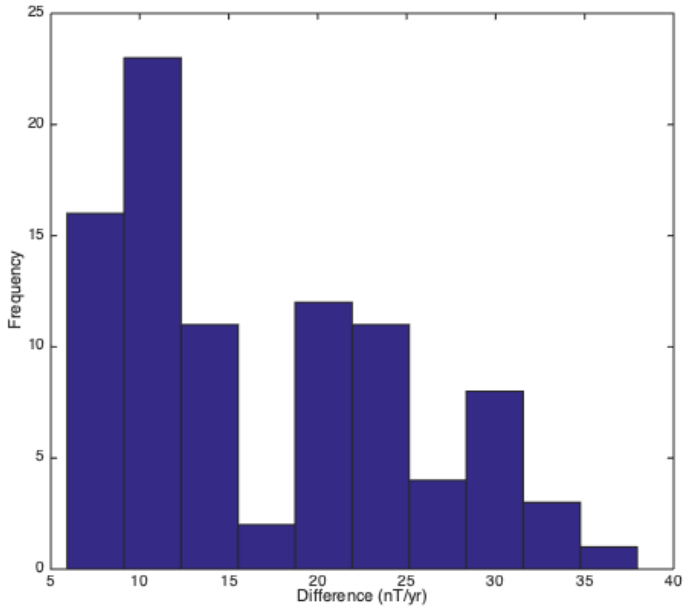


Figure 18: Histogram of RMS differences between SV candidates

	A	C	D	E	F	G	H	I	K	K	L	M	N	O	Mean	Median
A		18.1	8.3	7.6	10.4	10.7	19.5	20.7	8.5	11.6	28.2	12.6	12.0	24.4	8.9	9.0
C	18.1		11.3	19.2	12.6	14.6	10.6	26.6	13.7	12.5	32.7	9.8	12.1	23.9	11.8	10.9
D	8.3	11.3		10.1	6.8	8.3	12.2	21.3	6.7	6.8	28.8	5.9	6.8	21.9	3.9	2.4
E	7.6	19.2	10.1		12.0	12.9	20.7	21.0	11.1	13.4	29.6	14.7	13.3	24.6	10.7	10.8
F	10.4	12.6	6.8	12.0		10.9	13.6	22.4	8.6	9.3	31.5	8.8	9.4	23.0	7.7	6.9
G	10.7	14.6	8.3	12.9	10.9		15.6	19.3	10.2	8.6	28.8	9.7	10.7	25.0	7.8	7.5
H	19.5	10.6	12.2	20.7	13.6	15.6		26.3	15.1	12.0	32.6	9.3	11.4	24.4	12.4	11.6
I	20.7	26.6	21.3	21.0	22.4	19.3	26.3		21.8	19.7	32.7	21.7	22.0	29.2	19.0	20.5
K	8.5	13.7	6.7	11.1	8.6	10.2	15.1	21.8		9.1	28.9	9.1	9.2	23.3	6.8	6.7
K	11.6	12.5	6.8	13.4	9.3	8.6	12.0	19.7	9.1		28.0	6.0	8.8	24.6	6.0	5.3
L	28.2	32.7	28.8	29.6	31.5	28.8	32.6	32.7	28.9	28.0		28.5	29.1	38.0	26.7	27.9
M	12.6	9.8	5.9	14.7	8.8	9.7	9.3	21.7	9.1	6.0	28.5		6.8	23.6	5.9	4.6
N	12.0	12.1	6.8	13.3	9.4	10.7	11.4	22.0	9.2	8.8	29.1	6.8		21.0	6.2	6.3
O	24.4	23.9	21.9	24.6	23.0	25.0	24.4	29.2	23.3	24.6	38.0	23.6	21.0		21.0	22.4
Mean Diff	14.8	16.7	11.9	16.2	13.8	14.2	17.2	23.4	13.5	13.1	30.6	12.8	13.3	25.1	11.0	10.9

Table 3: RMS differences for the SV candidates

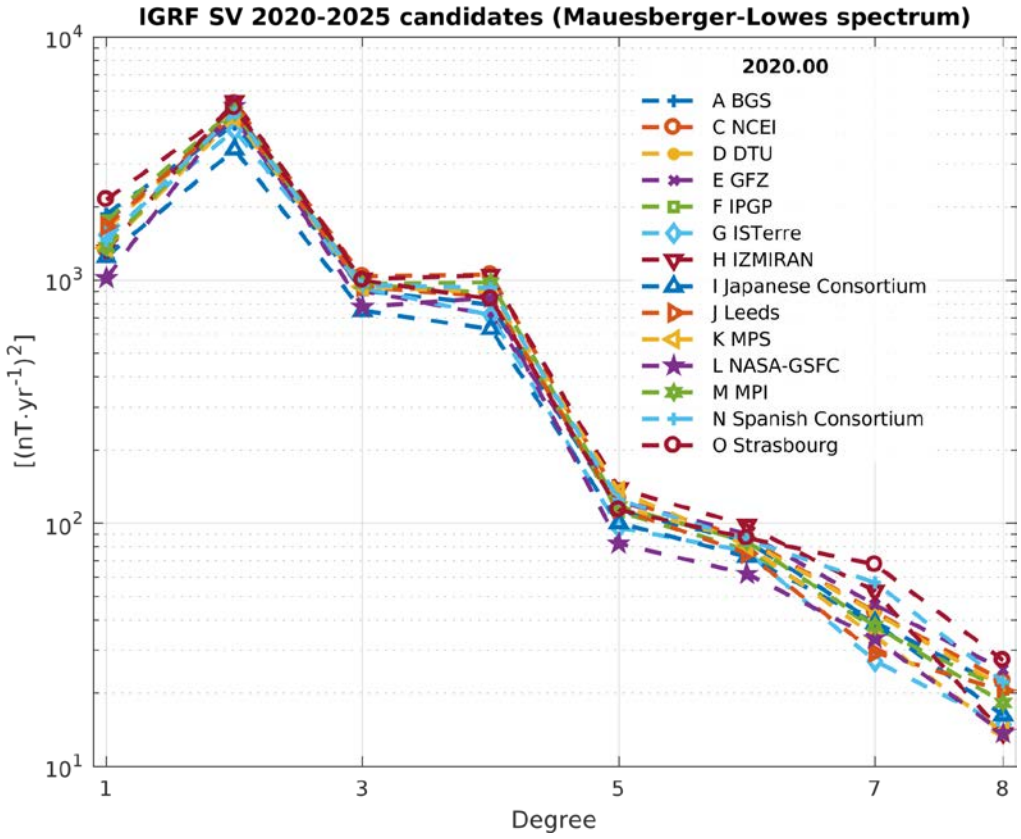


Figure 19: Lowes-Maursberger spectrum for the 14 SV 2020-2025 candidates

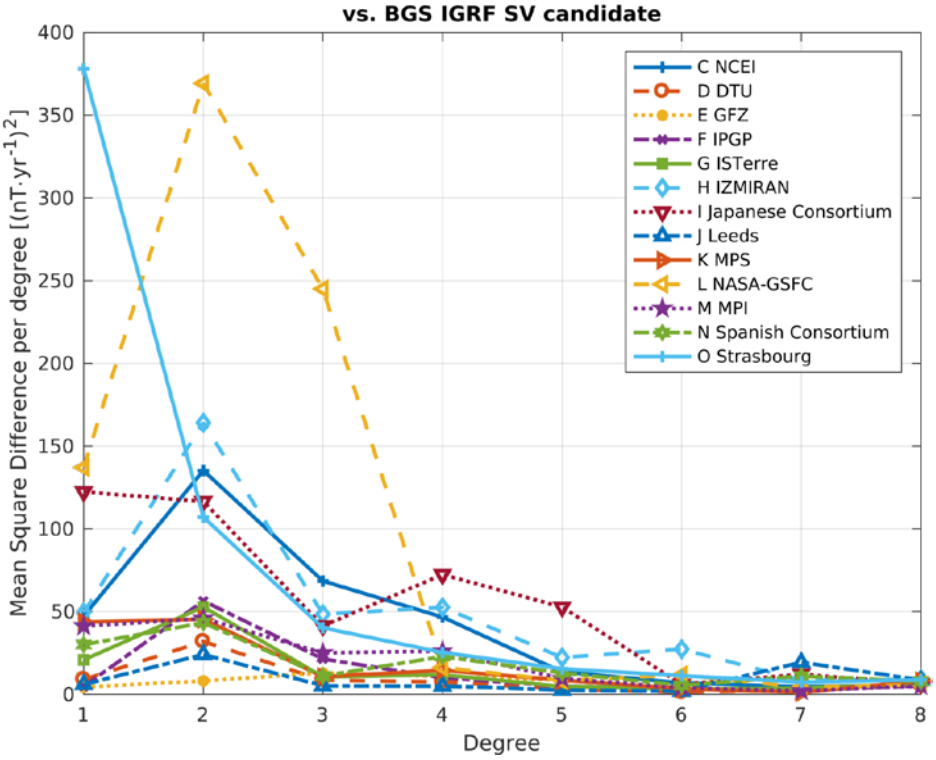


Figure 20: Mean square difference per degree between BGS and the other SV candidates

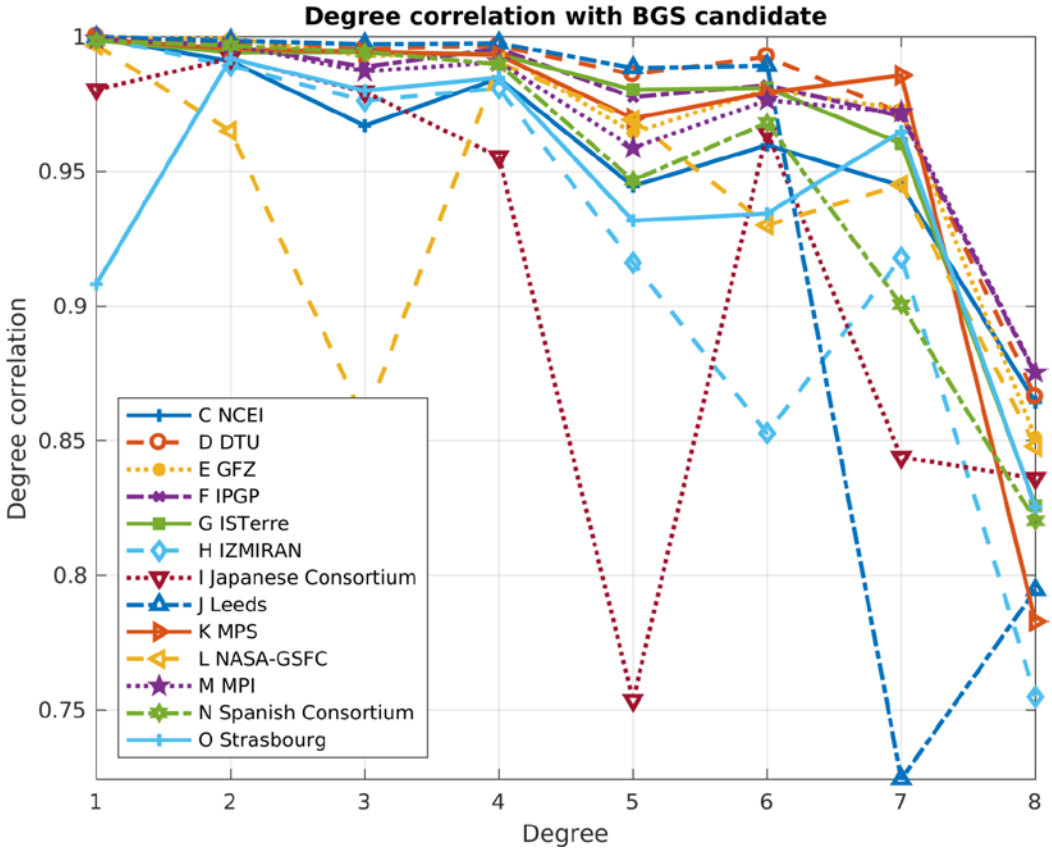


Figure 21: Degree correlation between BGS and the other SV candidates

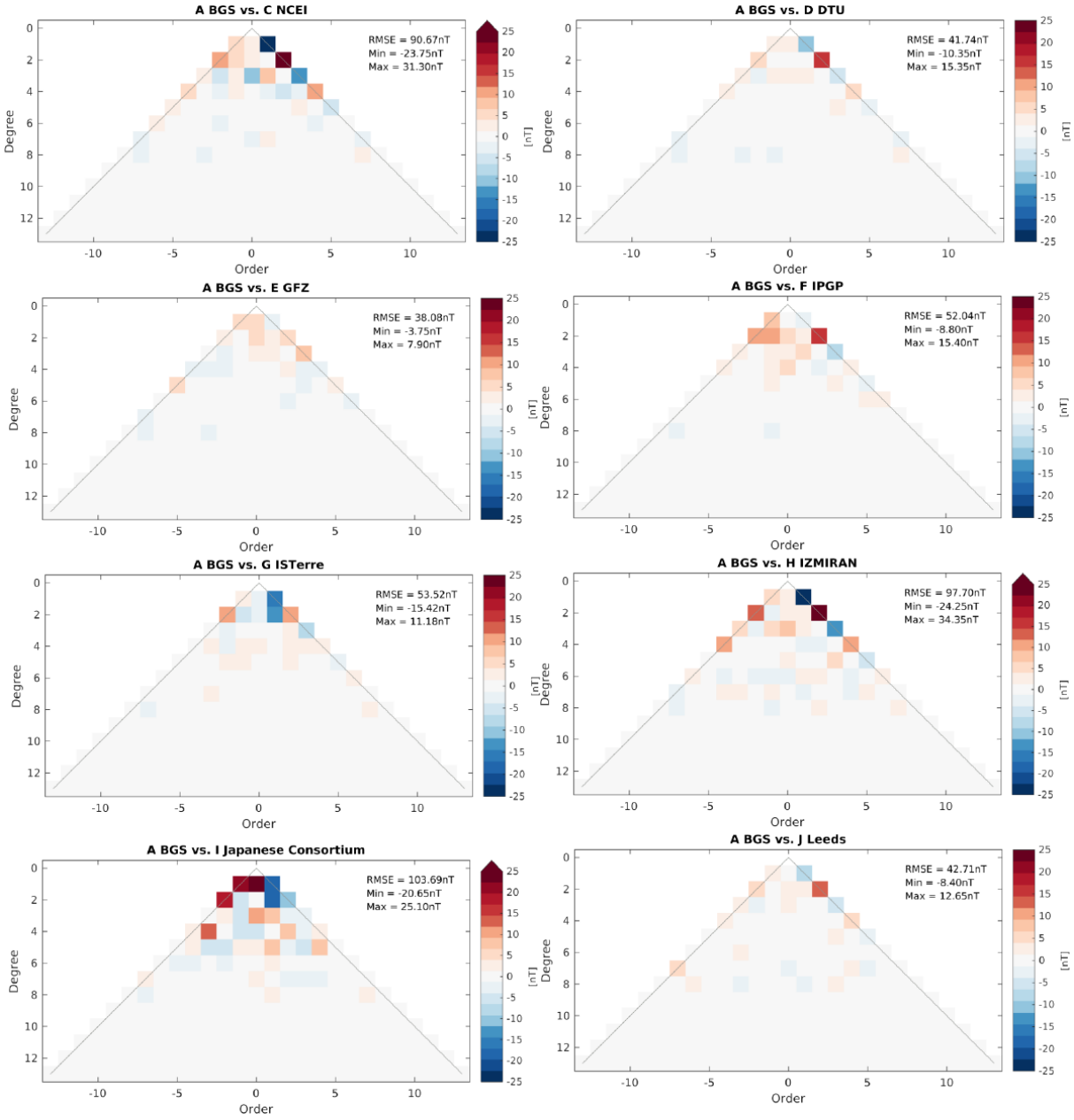


Figure 22: Triangle plots of coefficient differences between BGS and the A-J SV candidates

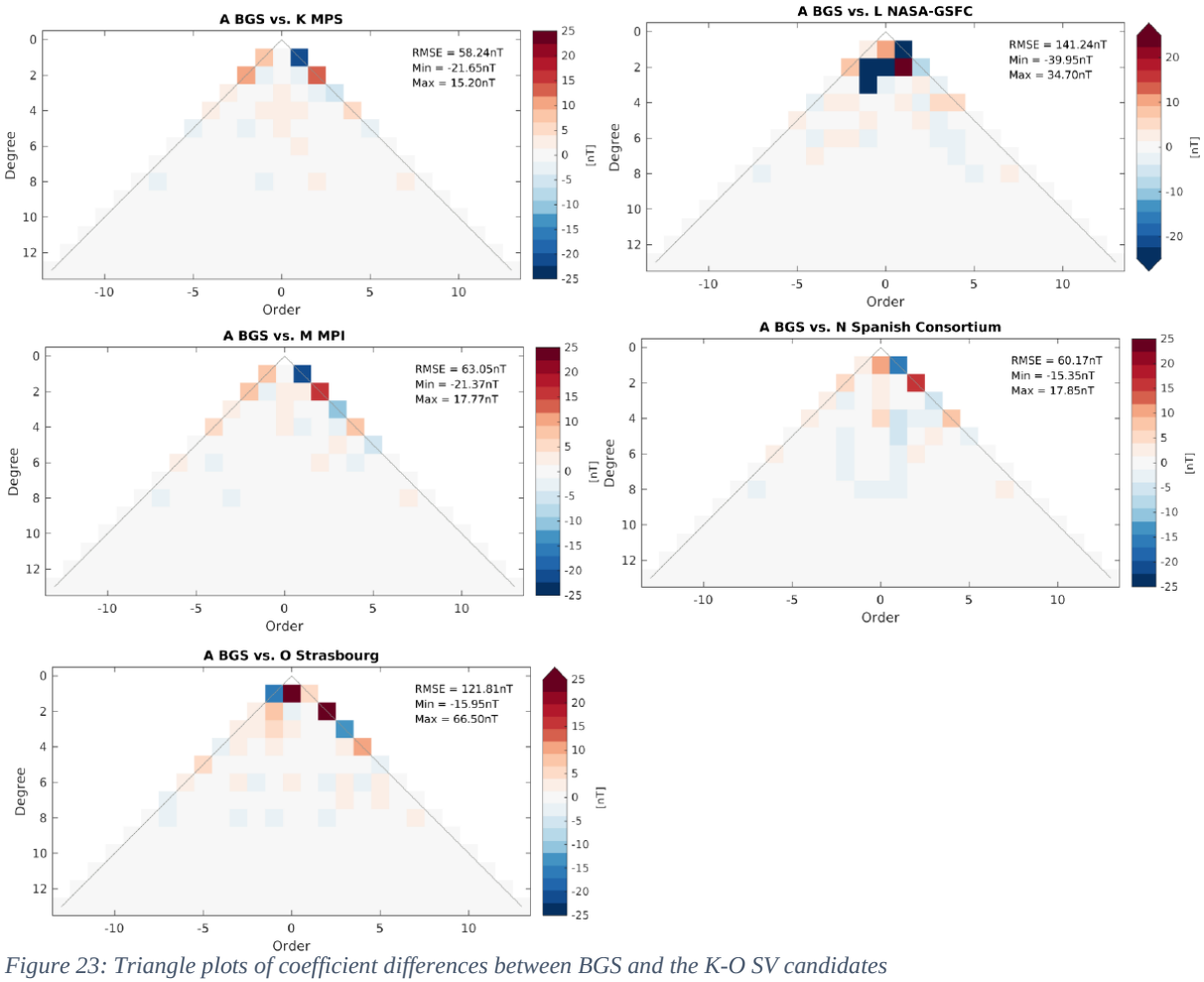


Figure 23: Triangle plots of coefficient differences between BGS and the K-O SV candidates

The triangle plots show the difference between the BGS candidate and the other model coefficients. Most show sine (positive order) differences, which may be a property of the BGS model rather than an issue with the other candidates.

There is a wide variation in the degree correlation plot too. Although most of the candidates show correlation above 0.95, some candidates disagree strongly (e.g. Leeds at degree 7).

The following spatial maps show the differences between the model candidates at 2025 (hence the units are in nT).

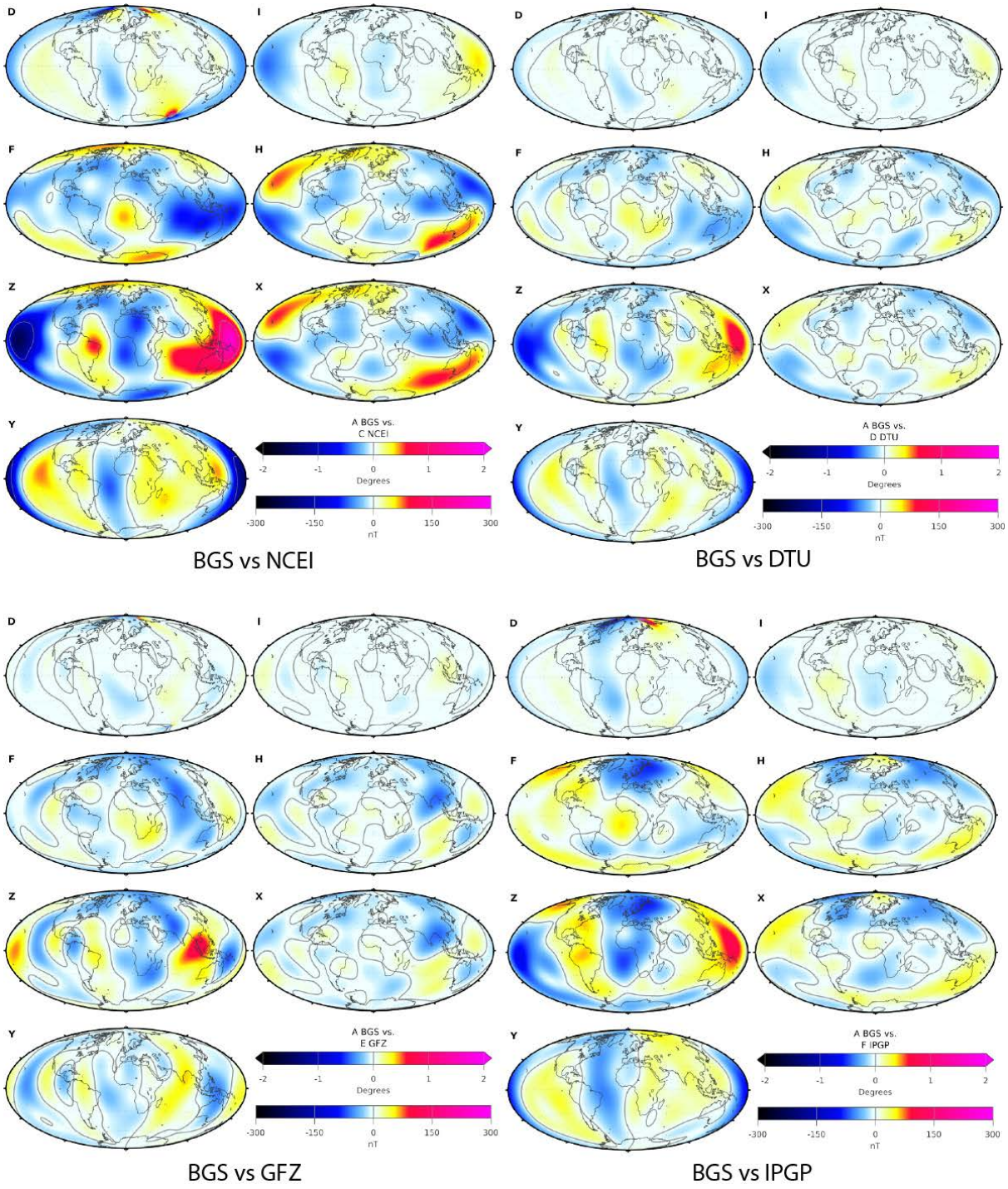


Figure 24: Map comparisons between the BGS SV candidate and NCEI, DTU, GFZ and IPGP

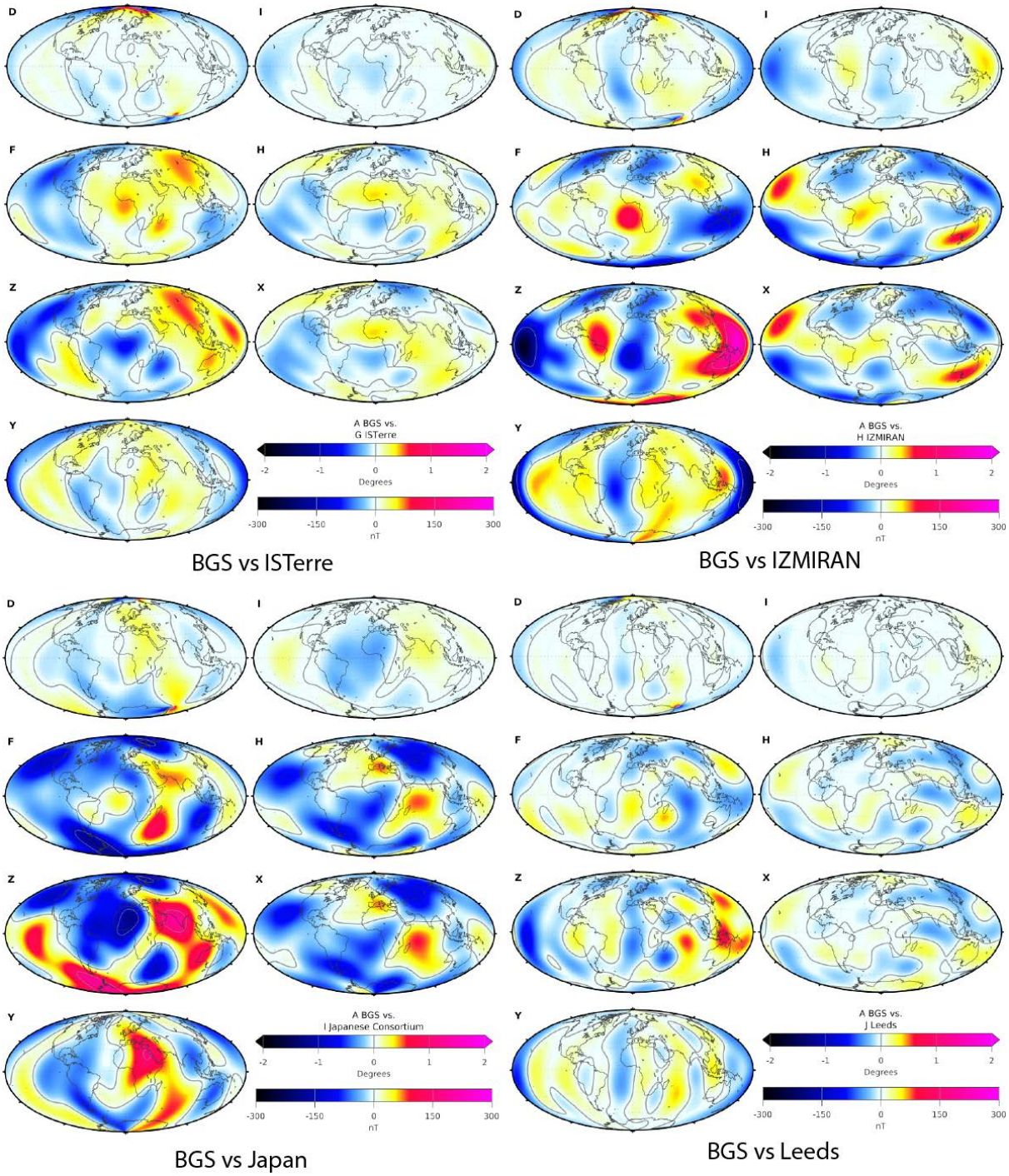


Figure 25: Map comparisons between the BGS SV candidate and ISTerre, IZMIRAN, Japan consortium and Leeds

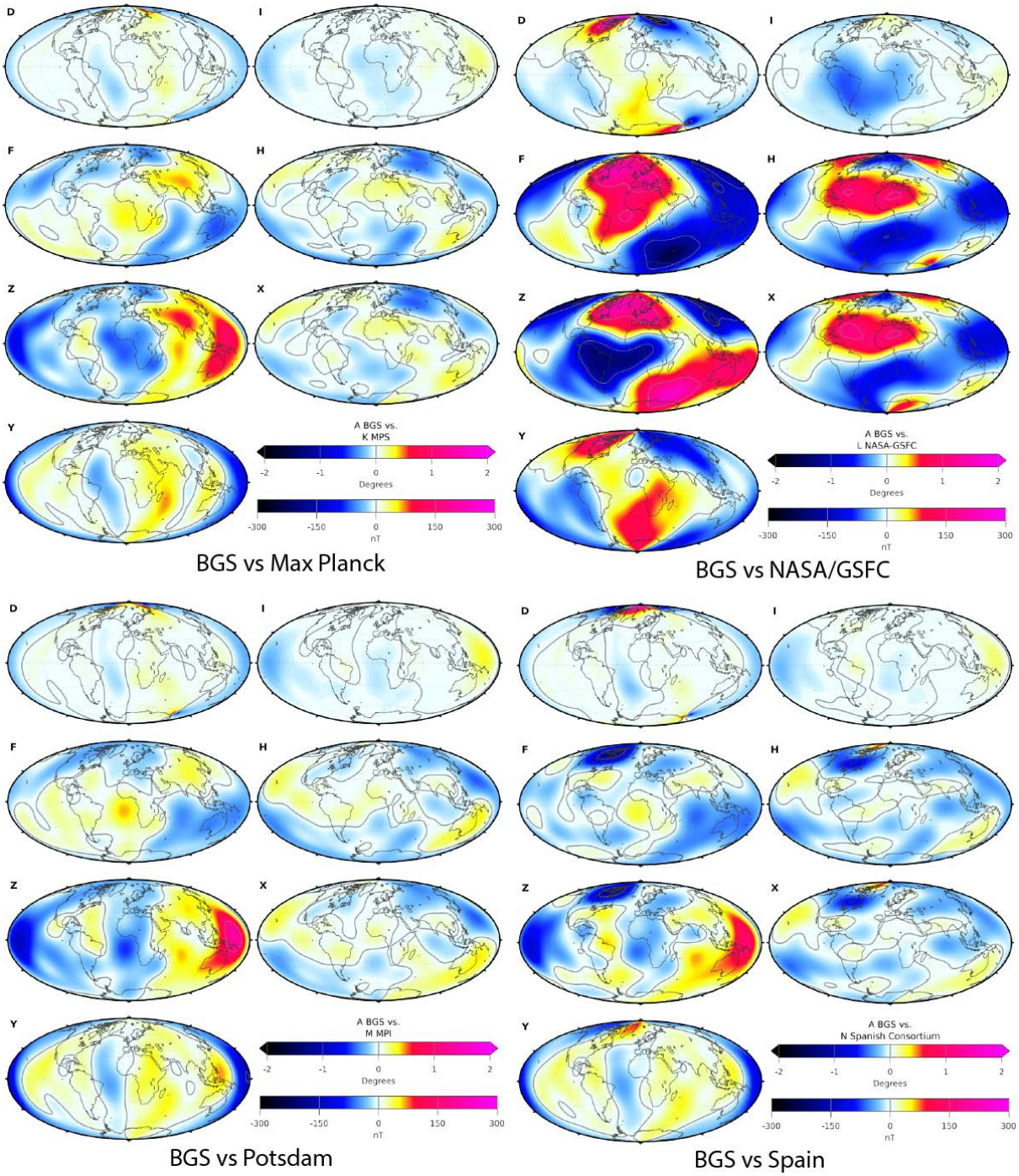


Figure 26: Map comparisons between the BGS SV candidate and Max Planck Institute, NASA/GSFC, Potsdam and Spanish consortium

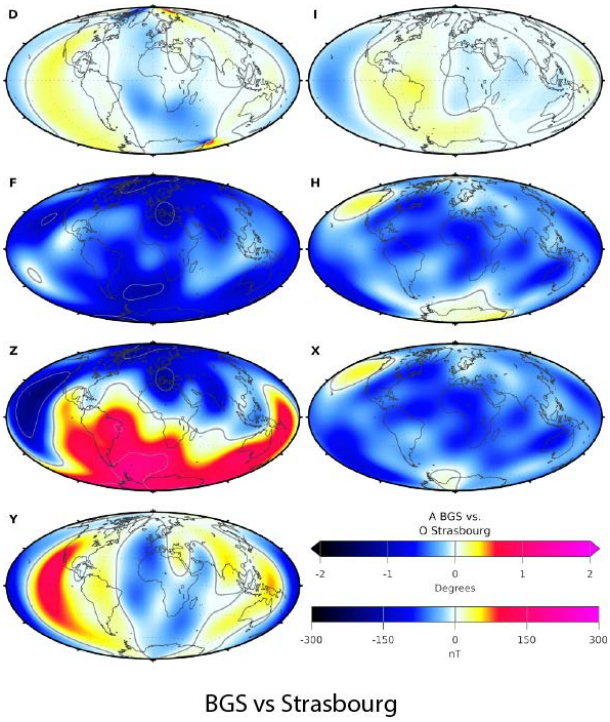


Figure 27: Map comparisons between the BGS SV candidate and Strasbourg

A common difference arises in the Pacific Ocean where the majority of candidates differ from the BGS candidate around the western and central Pacific in Z (i.e. CU/NCEI, DTU, IPGP, Potsdam, Spain). These are areas of rapid secular variation at present and thus the areas that show greatest difference depending on whether the magnitude of SV is predicted to be persistent or extrapolated from the last known values. The GFZ and DTU models are closest to the BGS model, in result but not SV prediction methodology. SV predictions cannot obviously be grouped by prediction method – there are several assimilated candidates, but the specifics of each method lead to significant differences in predictions.

The most different are the Japanese, NASA/GSFC and Strasbourg/EOST candidates. These show large, complex spatial variations.

Comparison to Recent Ground Observatory Data

As a final comparison, we plot the SV predictions from each candidate, as well as the mean and median candidates and compare them to monthly mean values from selected ground observatories. We chose the observatories in Figure 28 for a wide spatial distribution, including some (GUA and HON) where rapid changes in secular variation are observed. These observatories also have data available until August 2019. The full set of plots are in the attached PDF file, but we show the main field prediction and secular variation for Guam (GUA), Honolulu (HON) and Hermanus (HER) as examples.

Figure 29, Figure 30 and Figure 31 show the main field comparison to the hourly mean and monthly mean values at each observatory. Note the offset in the main field comparison is due to the crustal field bias at each location. The IGRF-12 model is also shown. The starting point for the main field are the coefficients of the mean of the candidate IGRF-13 models.

Figure 33, Figure 34 and Figure 35 show the comparison to the secular variation of the monthly mean values.

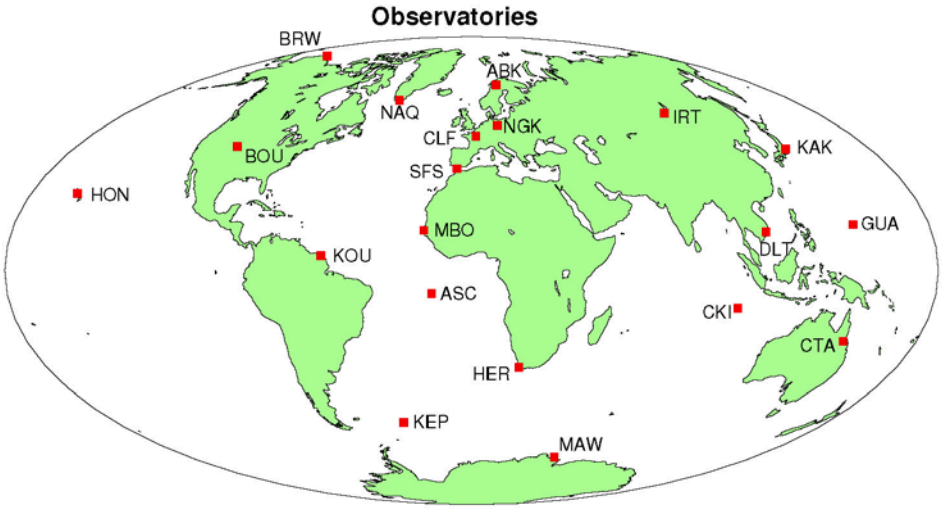


Figure 28: Location of observatories for SV comparison

Code	Lat	Long	Code	Lat	Long
ABK	68	19	HON	21	202
ASC	-8	346	IRT	52	104
BOU	40	255	KAK	36	140
BRW	71	203	KEP	-54	324
CKI	-12	97	KOU	5	307
CLF	48	2	MAW	-67	63
CTA	-20	146	MBO	14	343
DLT	12	108	NAQ	61	315
GUA	13	145	NGK	52	13
HER	34	19	SFS	36	354

Table 4: List and location of the 20 chosen observatories for SV comparison

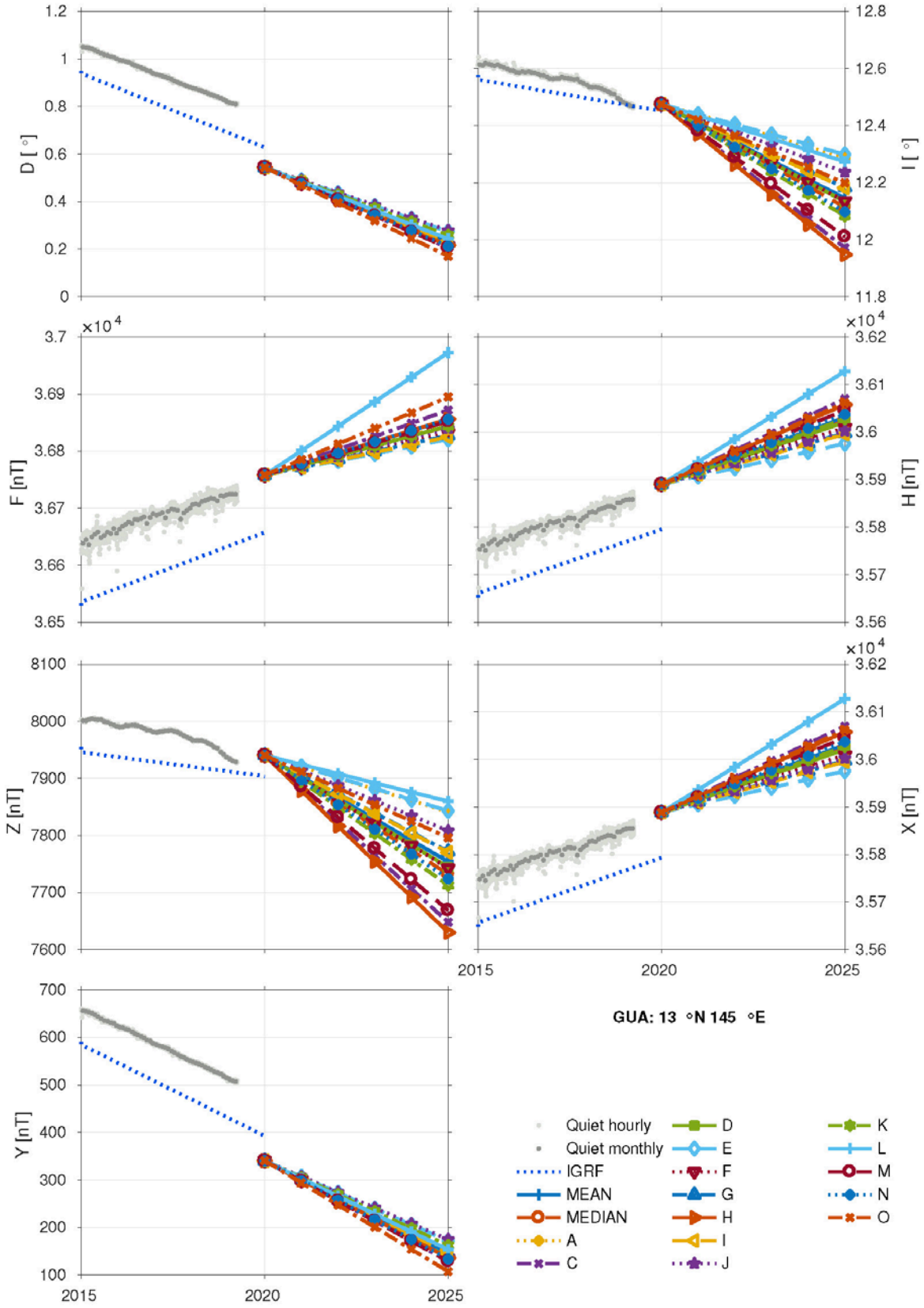


Figure 29: Main field comparison to all seven components for Guam for the SV candidates, mean and median model

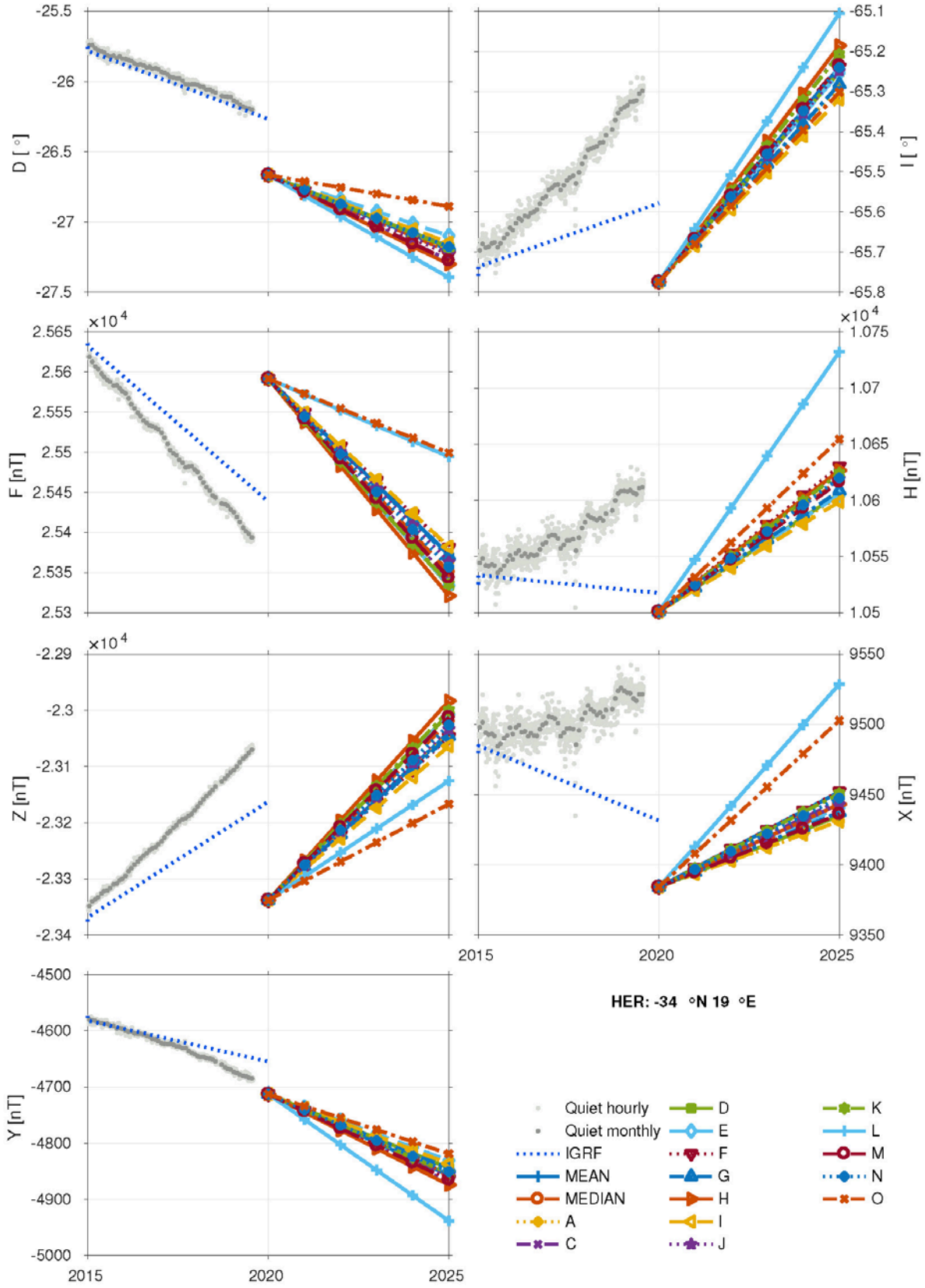


Figure 30: Main field comparison to all seven components for Hermanus for the SV candidates, mean and median model

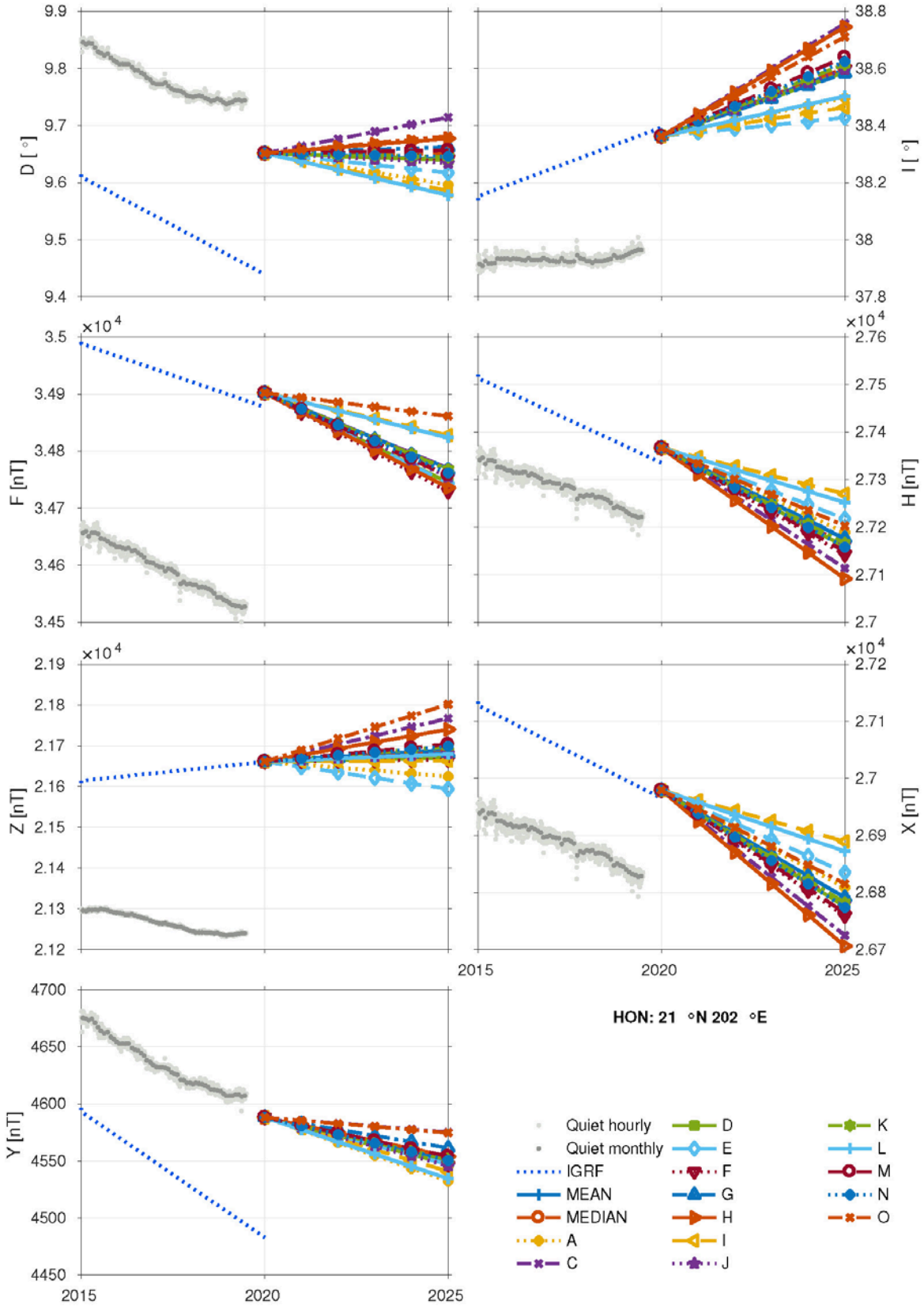


Figure 31: Main field comparison to all seven components for Honolulu for the SV candidates, mean and median model

Figure 32: Main field comparison to all seven components for Honolulu for the SV candidates, mean and median model

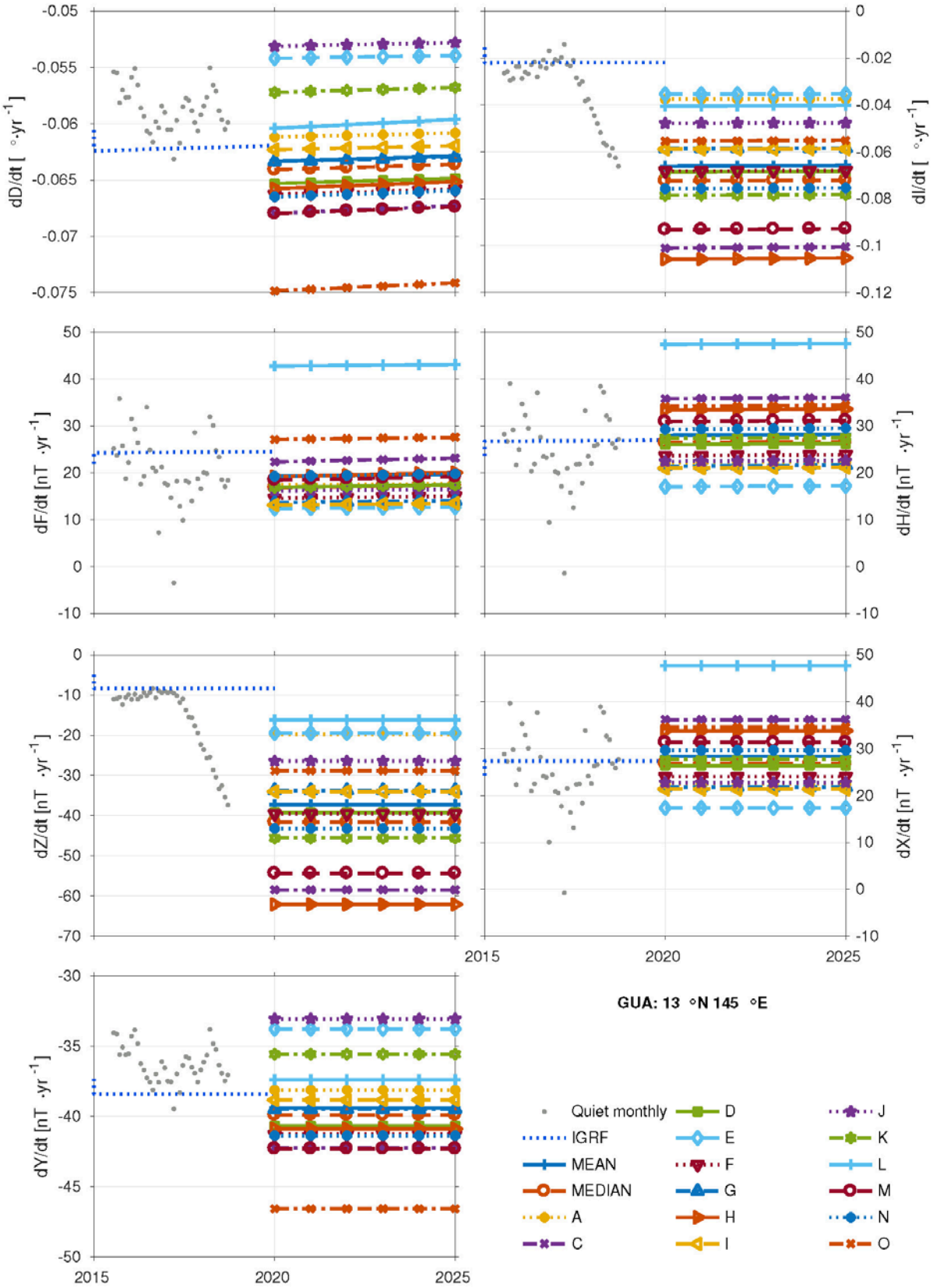


Figure 33: Secular variation comparison to all seven components for Guam for the SV candidates, mean and median model

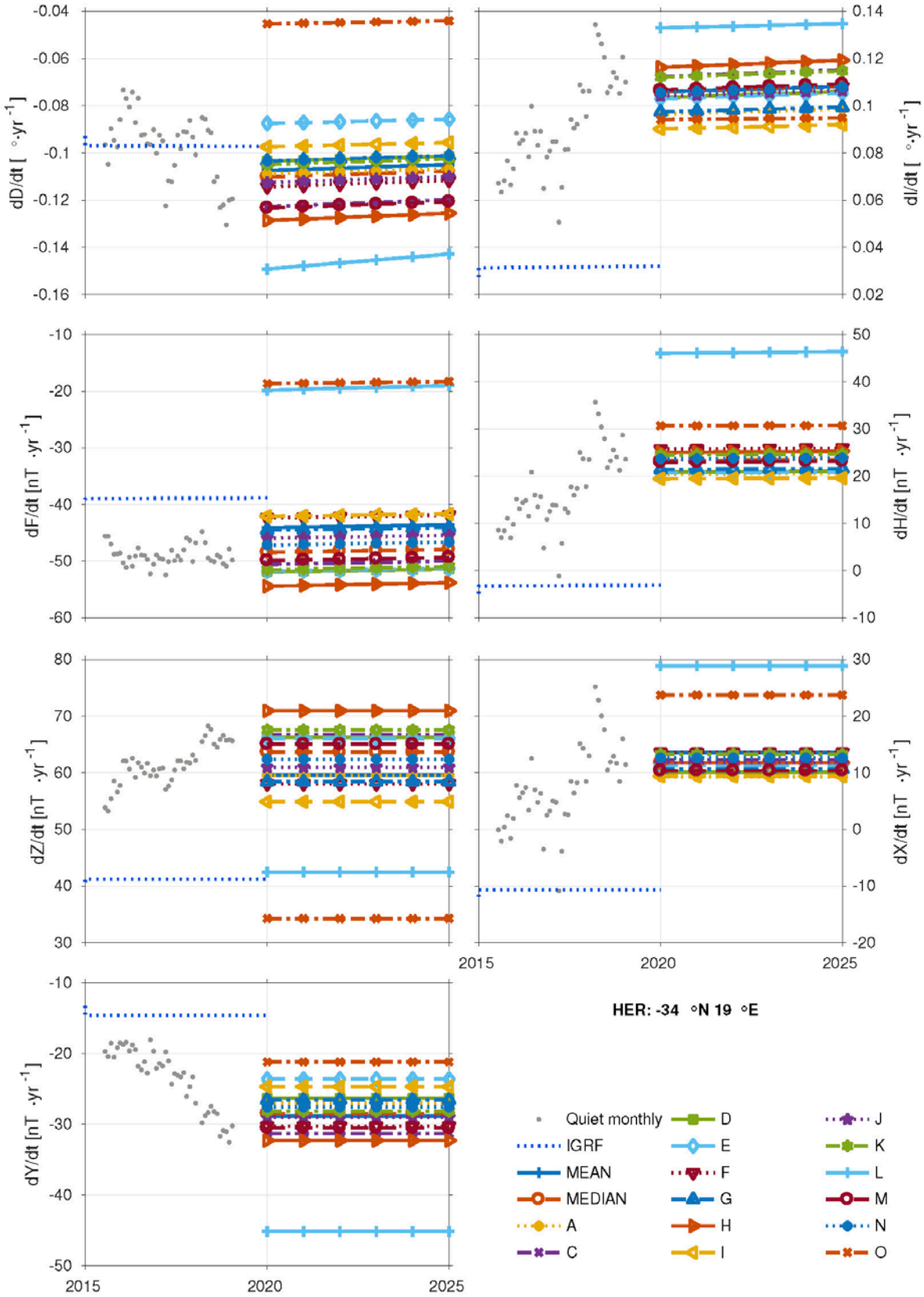


Figure 34: Secular variation comparison to all seven components for Hermanus for the SV candidates, mean and median model

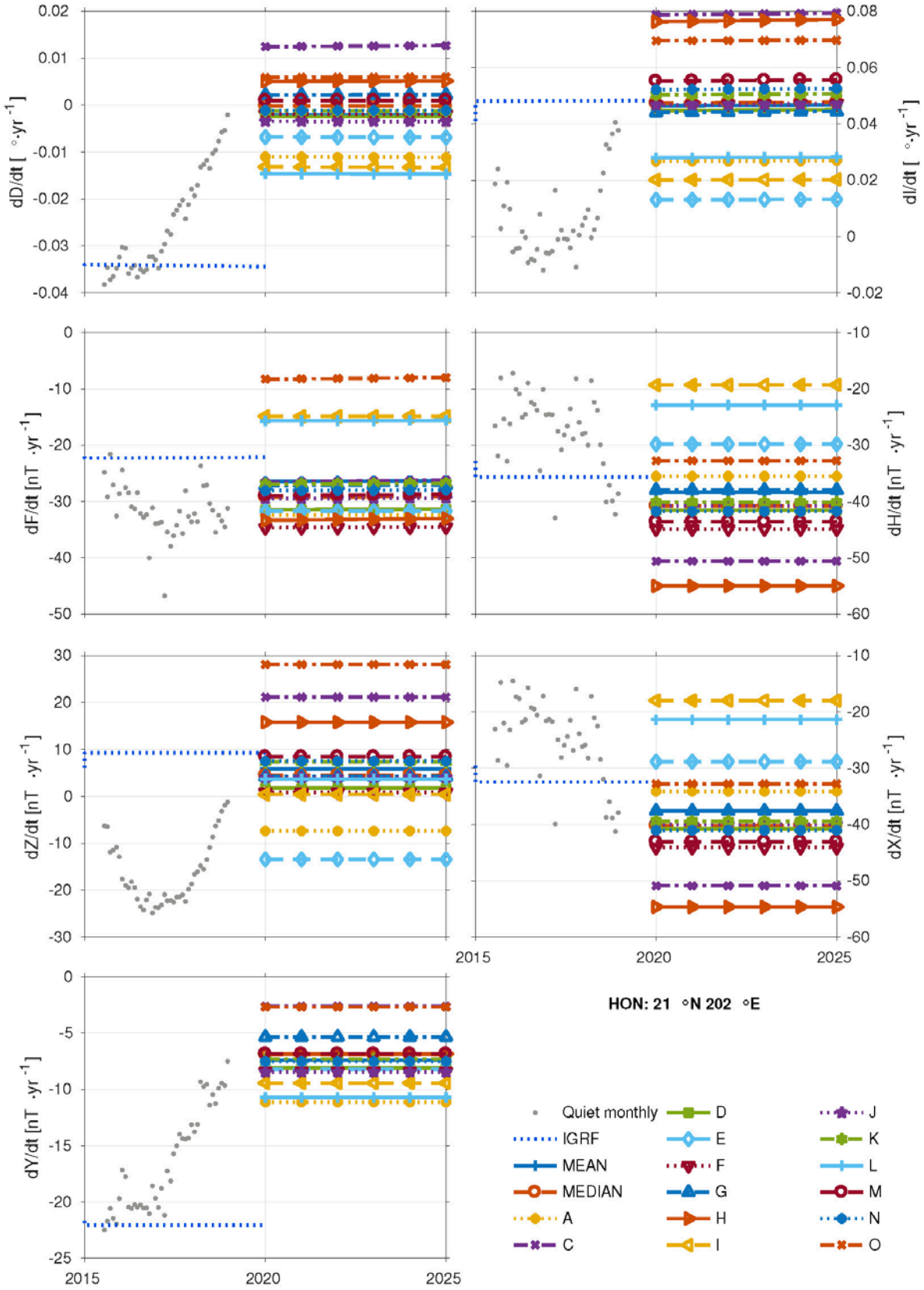


Figure 35: Secular variation comparison to all seven components for Honolulu for the SV candidates, mean and median model

From the main field and secular variation comparisons, it is clear there is a wide variation in the prediction of how the magnetic field will change between 2020 and 2025. Again, the IZMIRAN, Japan and NASA/GSFC models are most different from the other candidates.

Interestingly there is not much difference between the mean and median models and, in general, they appear to be a good description of secular variation trends.

Suggested methods for creating the IGRF-13 models

DGRF-2015

Given the excellent data available from the Swarm mission from November 2013 onwards, all models have produced a very close match to each other. Much of the differences below 5 nT RMS can be attributed to modelling and data selection methodology. Only the ISTerre and IZMIRAN models are slightly larger than 5 nT RMS difference with the other candidates.

In this case, we would recommend using the simple arithmetic mean model.

IGRF-2020

The differences in the candidate models arise from the extrapolation of the main field models from the end point of the dataset (mid-2019) to 2020.0. This allow slightly larger variation to occur.

Again, there is no clear ‘clustering’ of candidates together which suggests that a simple mean or median model may well be the best methodology to use. A more complicated weighting scheme using the Mean square difference per degree between the mean model and candidates could be envisaged, which could weight different coefficients. However, the outcome would probably be quite close to the median model in any case.

We would recommend the median model.

SV-2020-2025

There is a large variation of the methodologies and approaches used. The models most different are the NASA and Japanese model. However, the direct comparisons of the mean and median models to the ground-based observatory time series suggests that these average models are capturing the general trends of secular variation relatively well. Indeed they overlap strongly in most locations.

Again, it may be easiest to use the arithmetic mean or median models to model the secular variation.